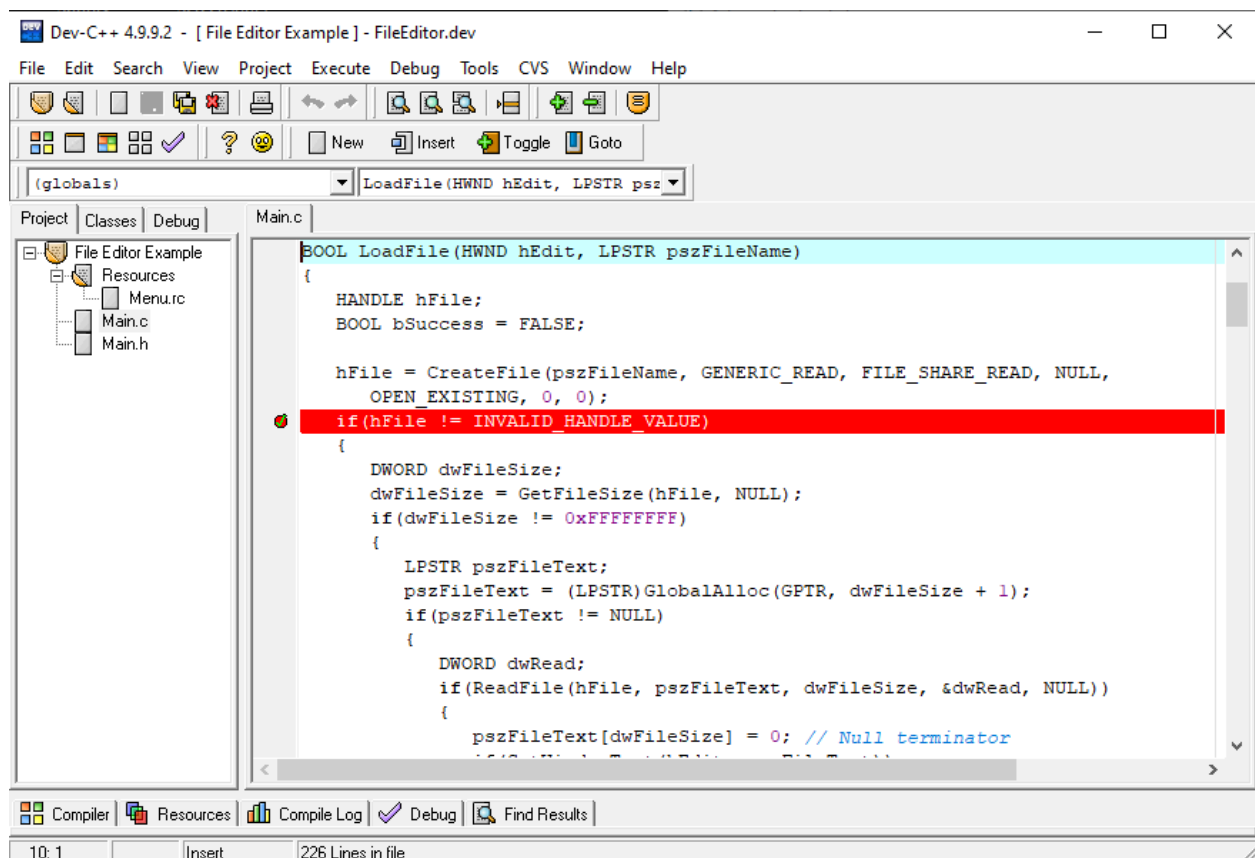


DEV C++

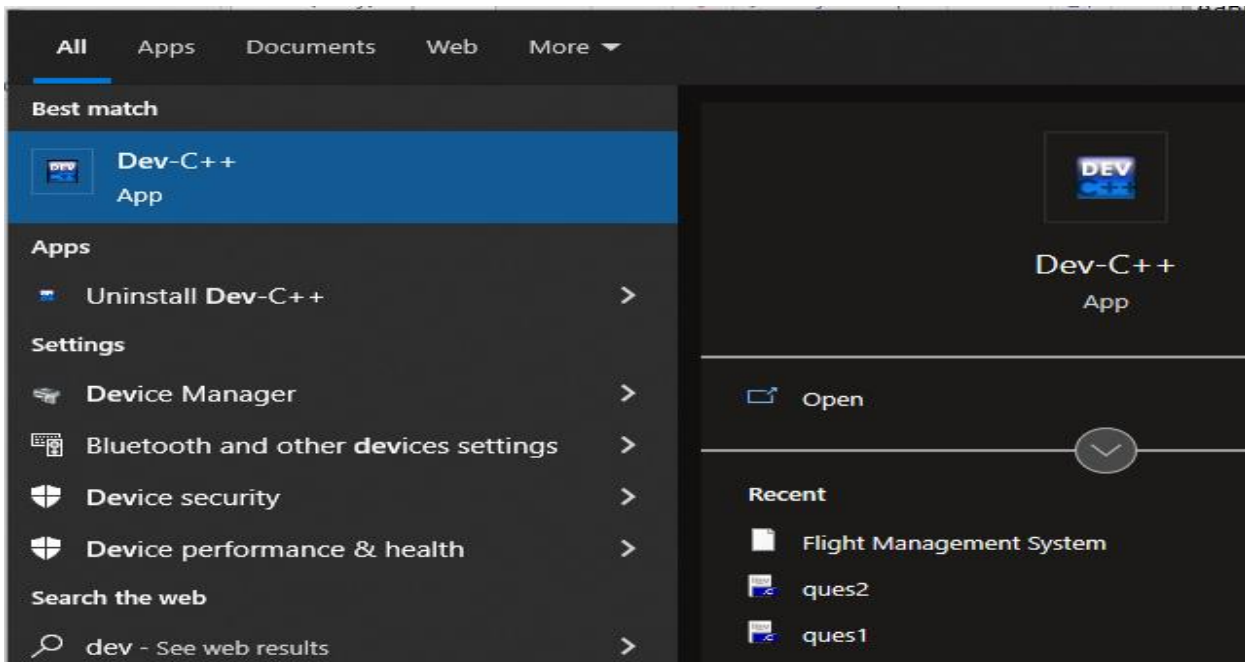
Installation and Getting Started under Windows OS

Dev-C++ is a full-featured C and C++ Integrated Development Environment (IDE) for Windows platforms. Millions of developers, students and researchers use Dev-C++ since the first version was released in 1998. It has been featured in dozens of C++ and scientific books and remains one of the favorite learning tool among universities & schools worldwide.

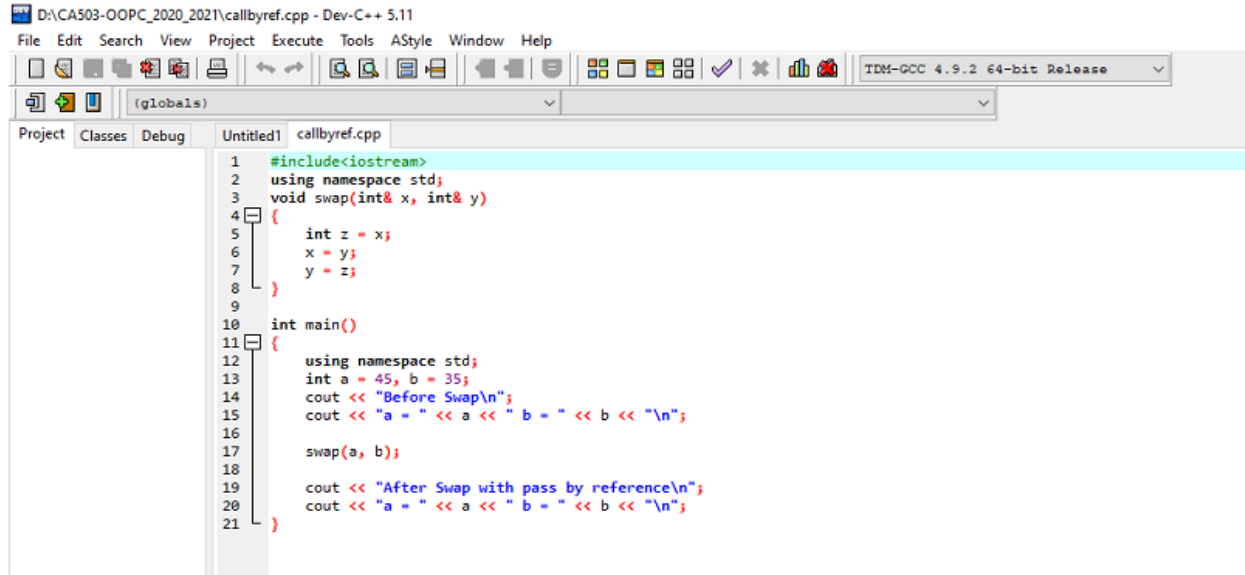
- 1 • Go to <https://www.bloodshed.net/>
- 2 • Click on [Download Original Dev-c++ for Windows](#)
- 3 • Click on [install Dev C++ for the first time](#)
- 4



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Select Dev-c++ from start menu,it open Dev-c++ editor



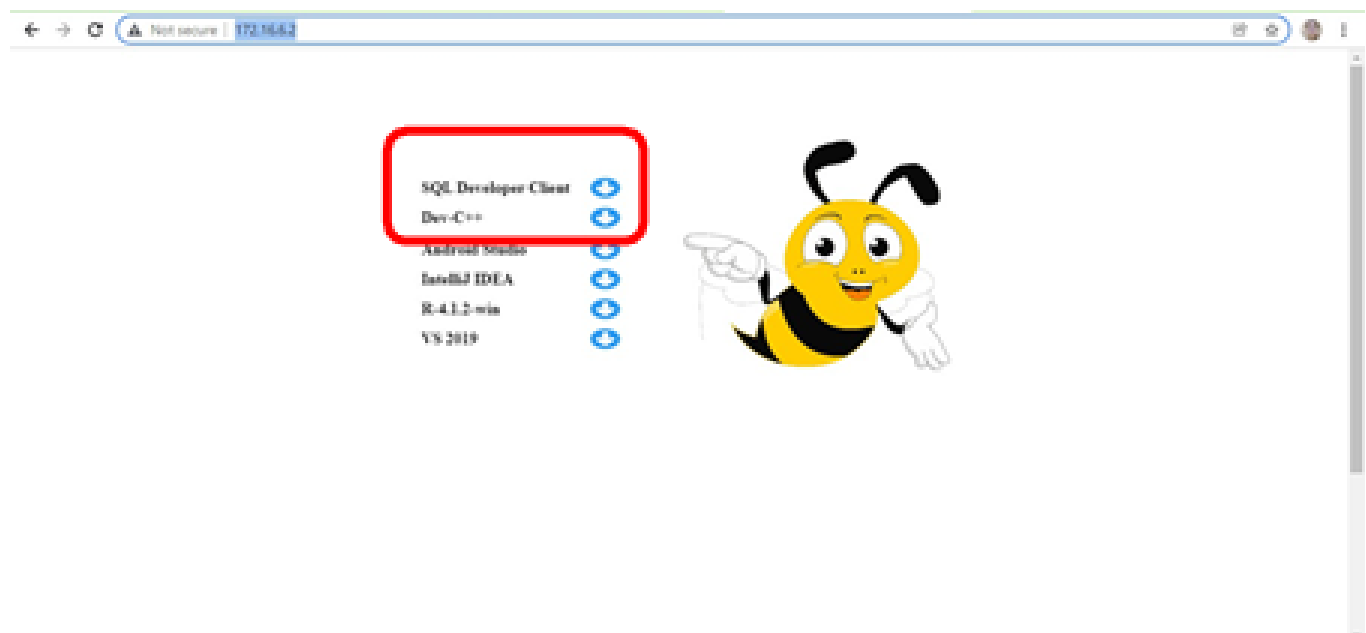
Write your code in Editor, save it and Select Execute->Compile and Execute->Run (F10)

Objectives

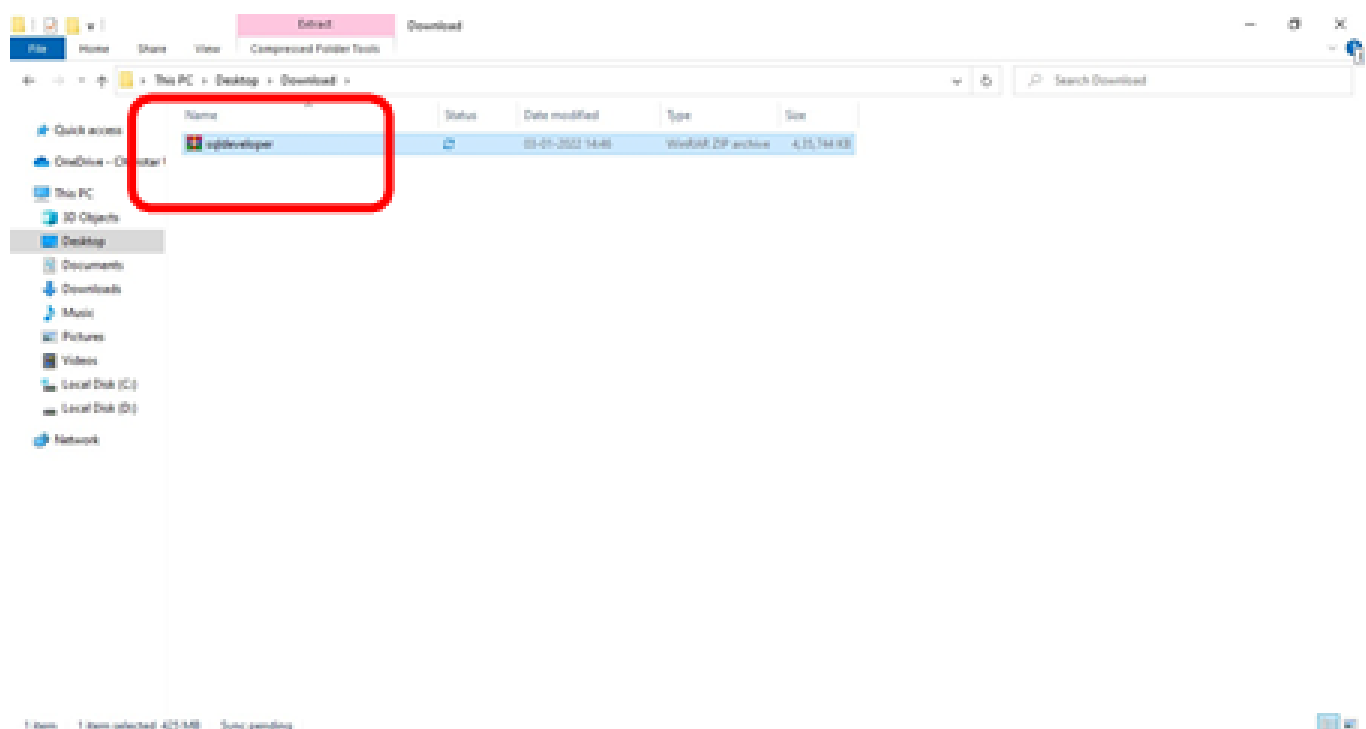
- To make students acquaint with installation of Database Client Software.

Step: 1 Go to <http://172.16.6.2/>

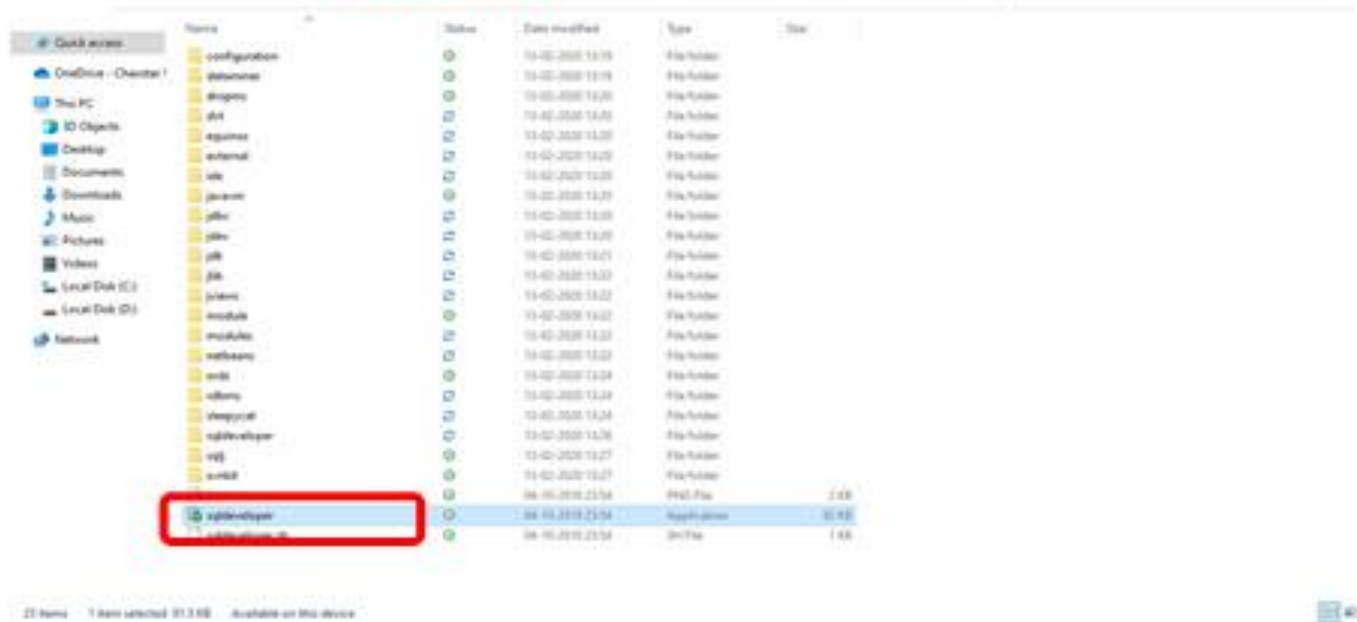
Download SQL Developer Client



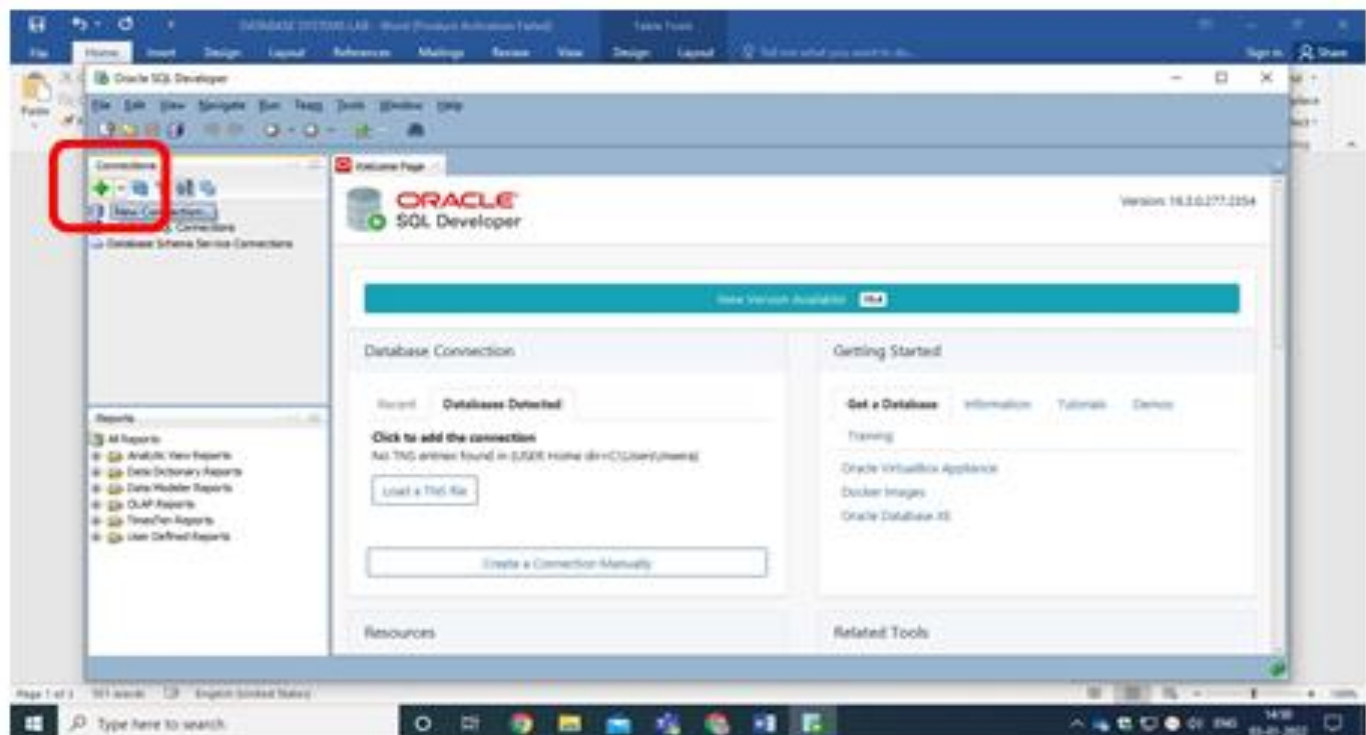
This will download the zip with name sqldeveloper. Unzip it.



Step: 2 Double click on sqldeveloper Application

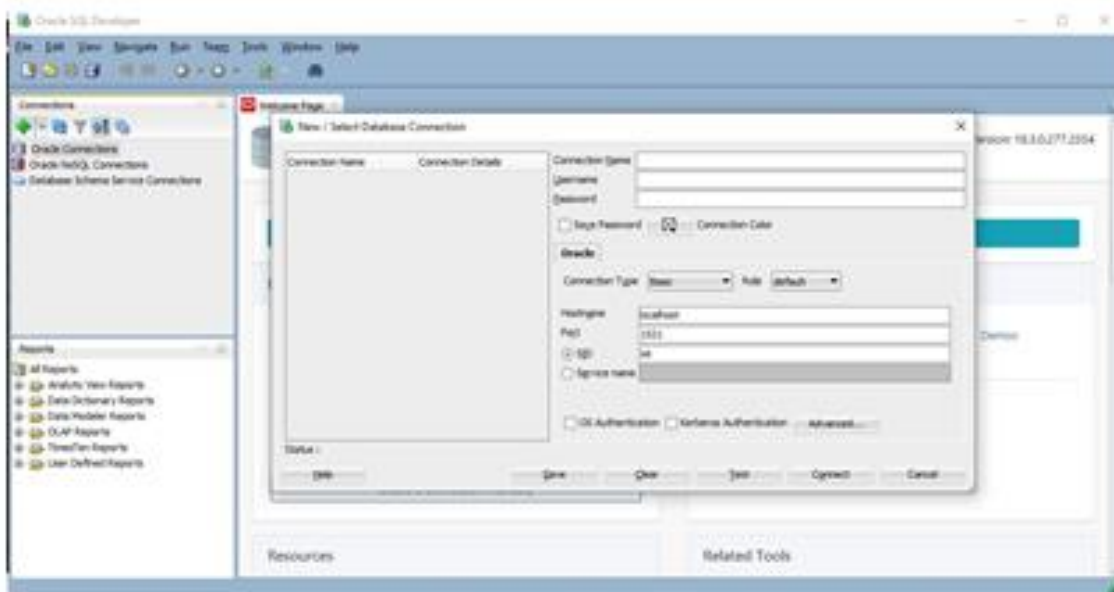


It will open the Oracle SQL Developer.



Step: 3 Click on + Sign, the following screen will appear.

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Step: 4 Give Proper Connection Name for example CMPICA

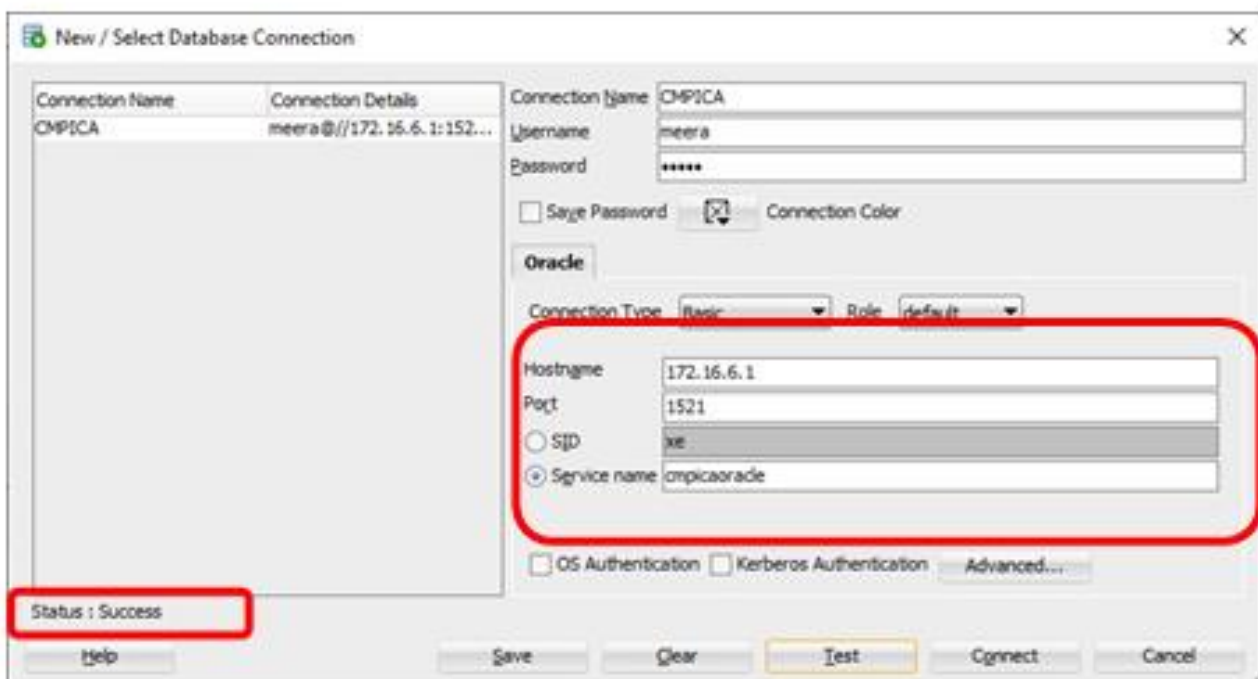
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

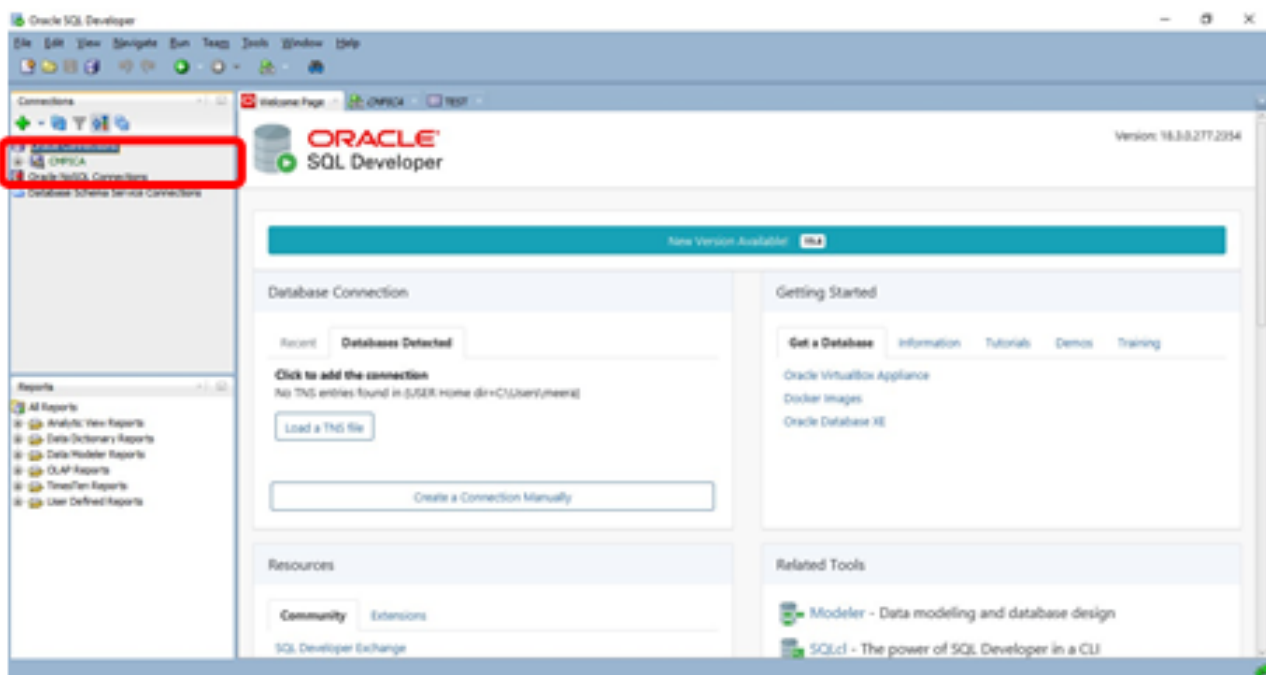
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **SUCCESS** should get displayed.

Now click on Connect Button.

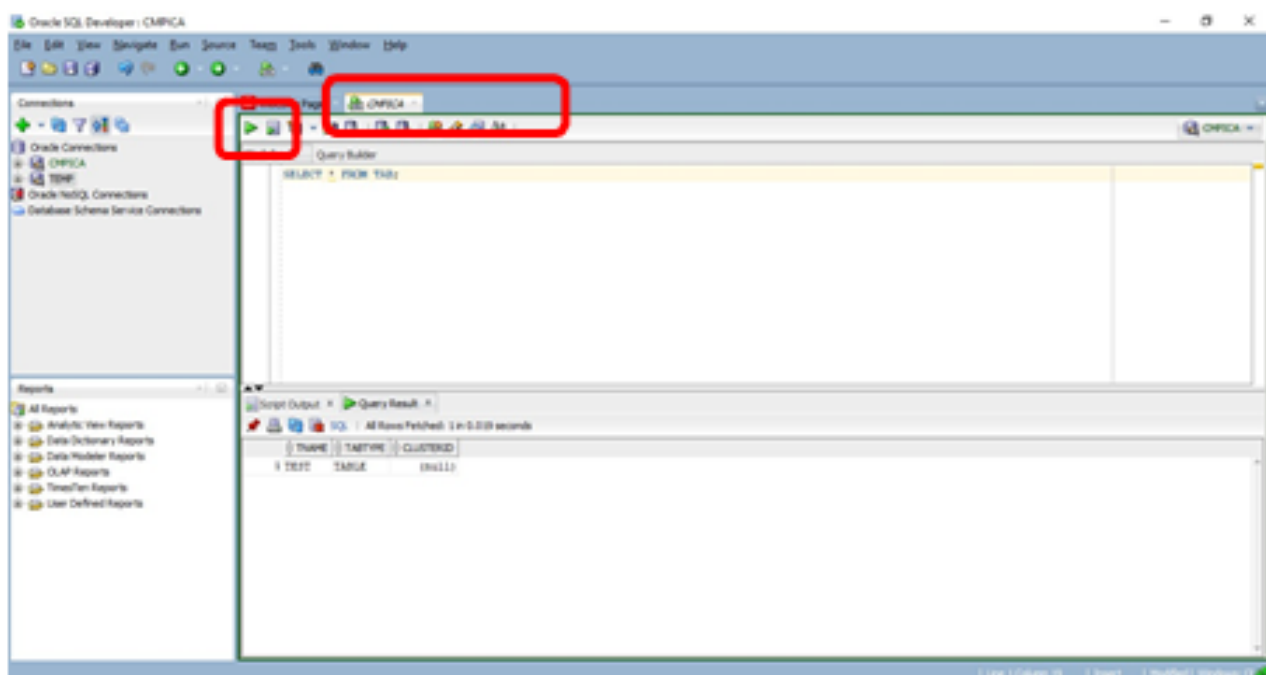


Your newly created connection should get displayed on the left panel.

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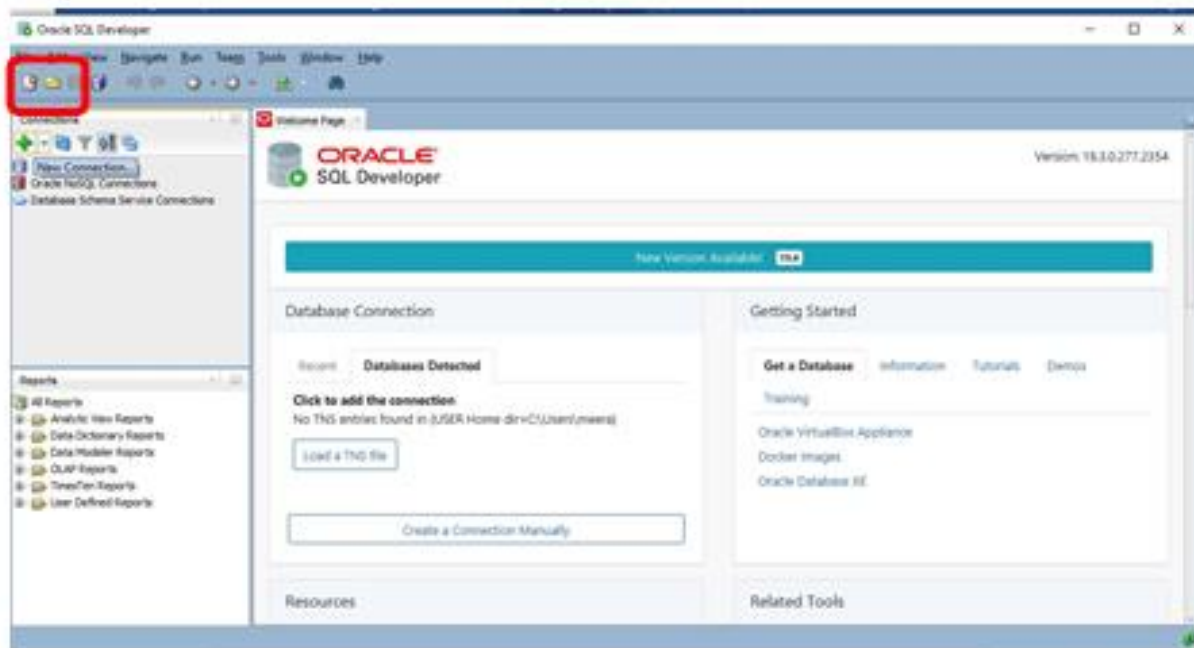
Step 5 On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter.



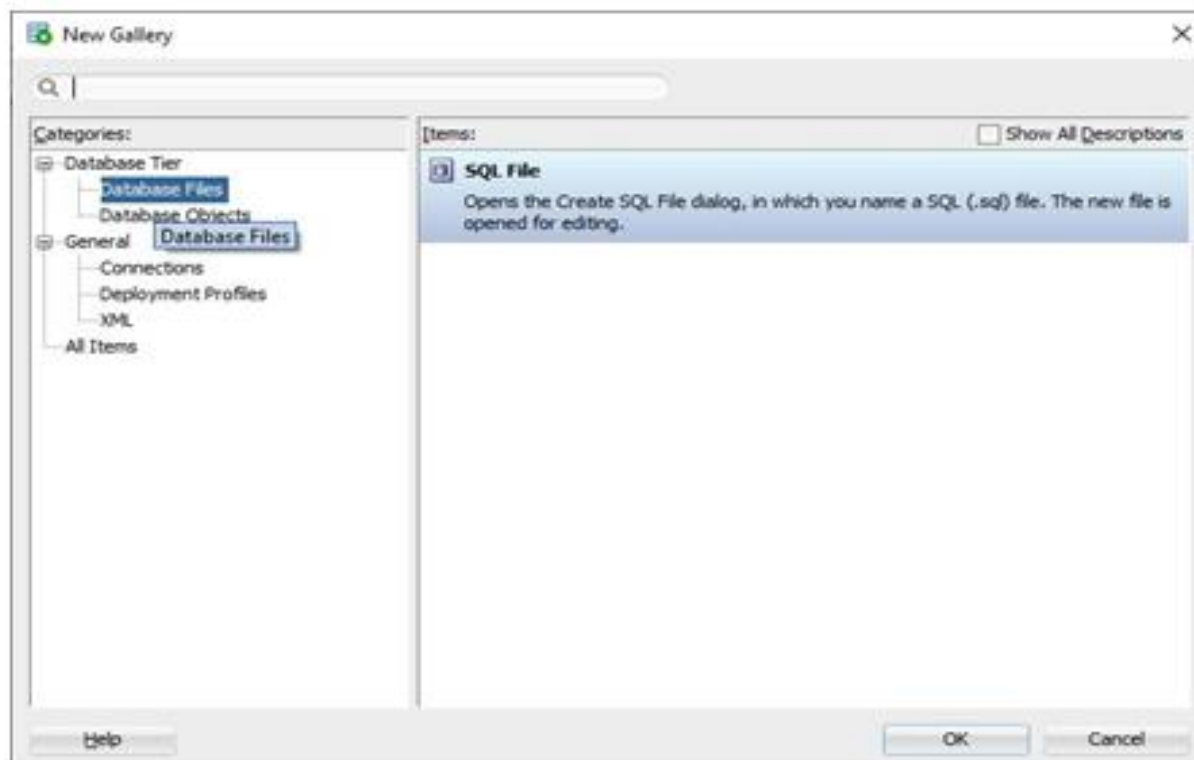
Step 6 IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

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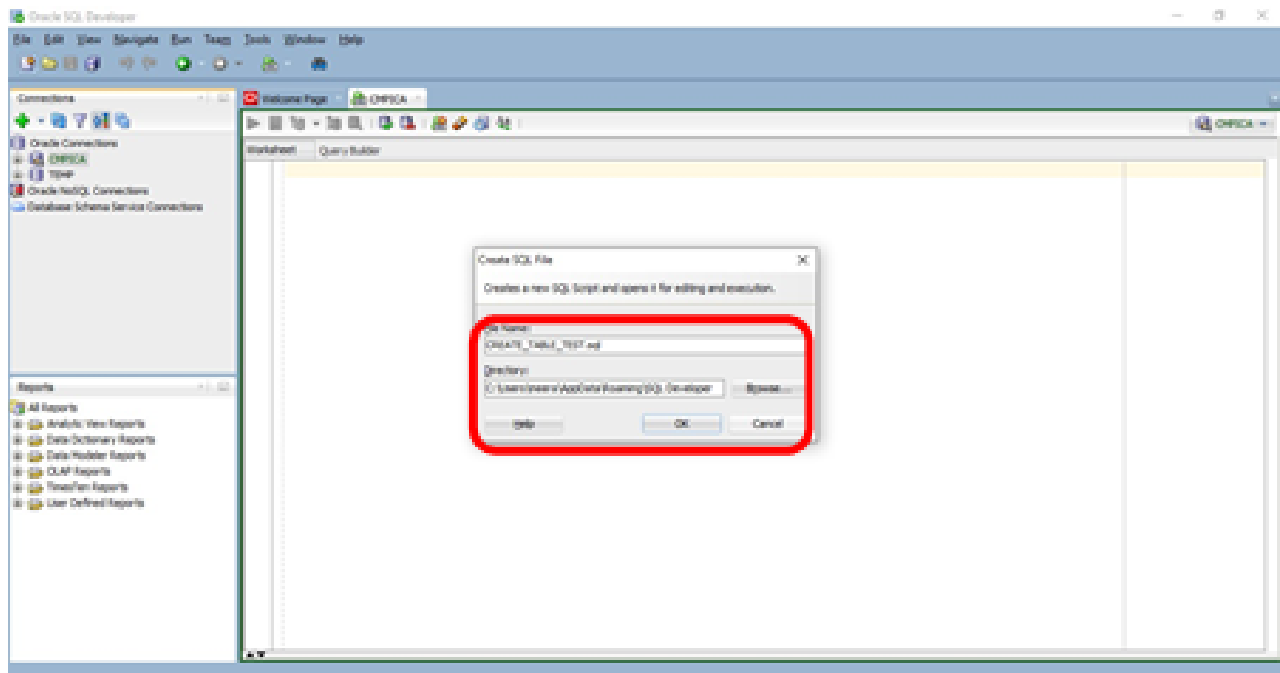


The following screen will appear.



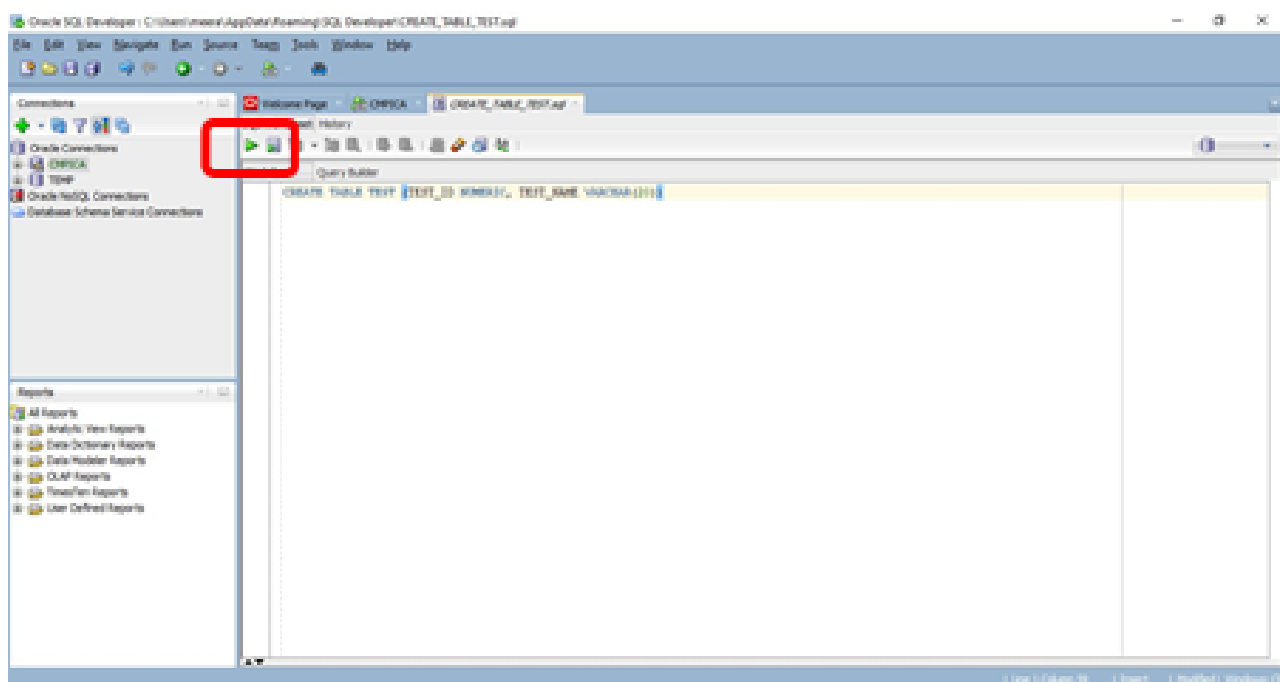
Give proper file name and save to the proper location. [IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]

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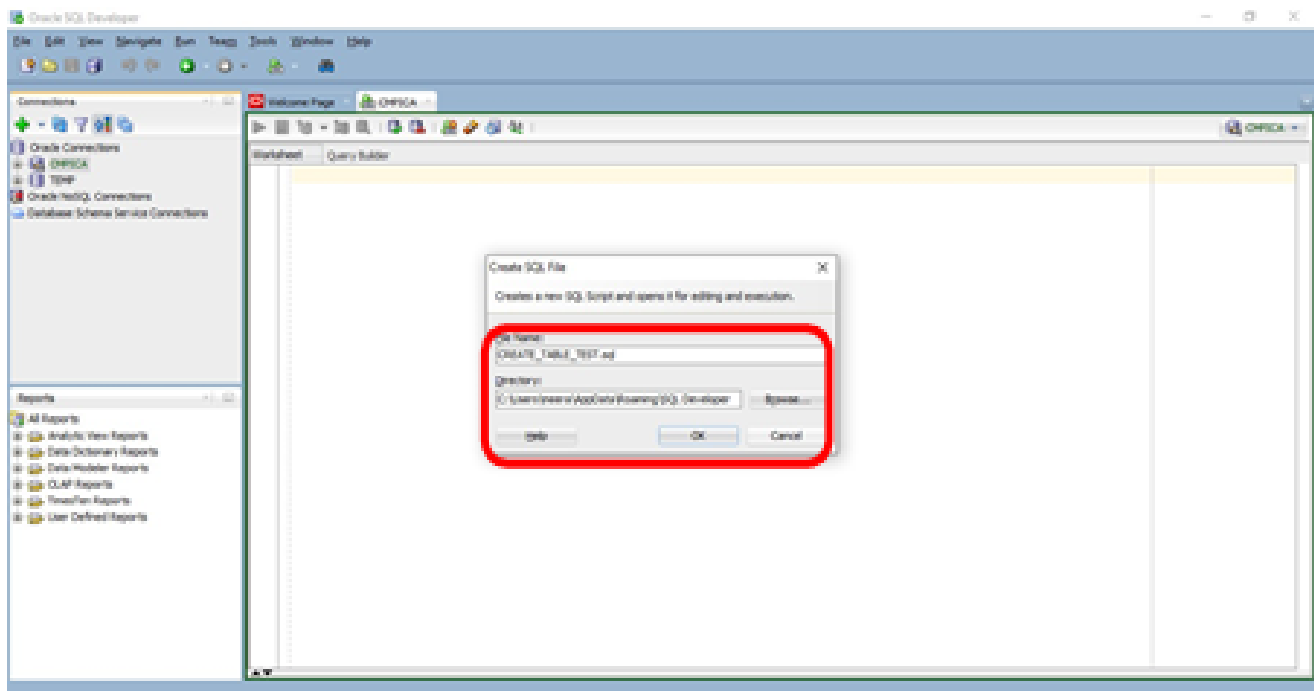
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.



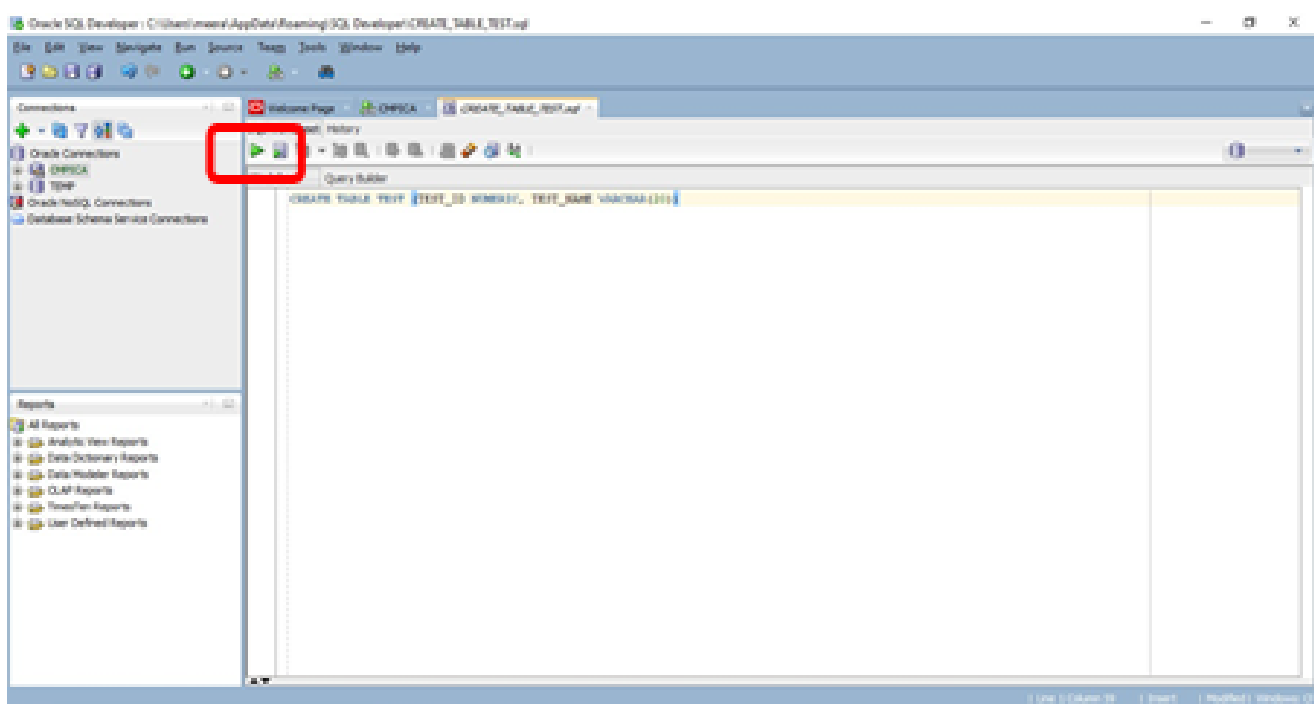
Select the appropriate connection and press OK.

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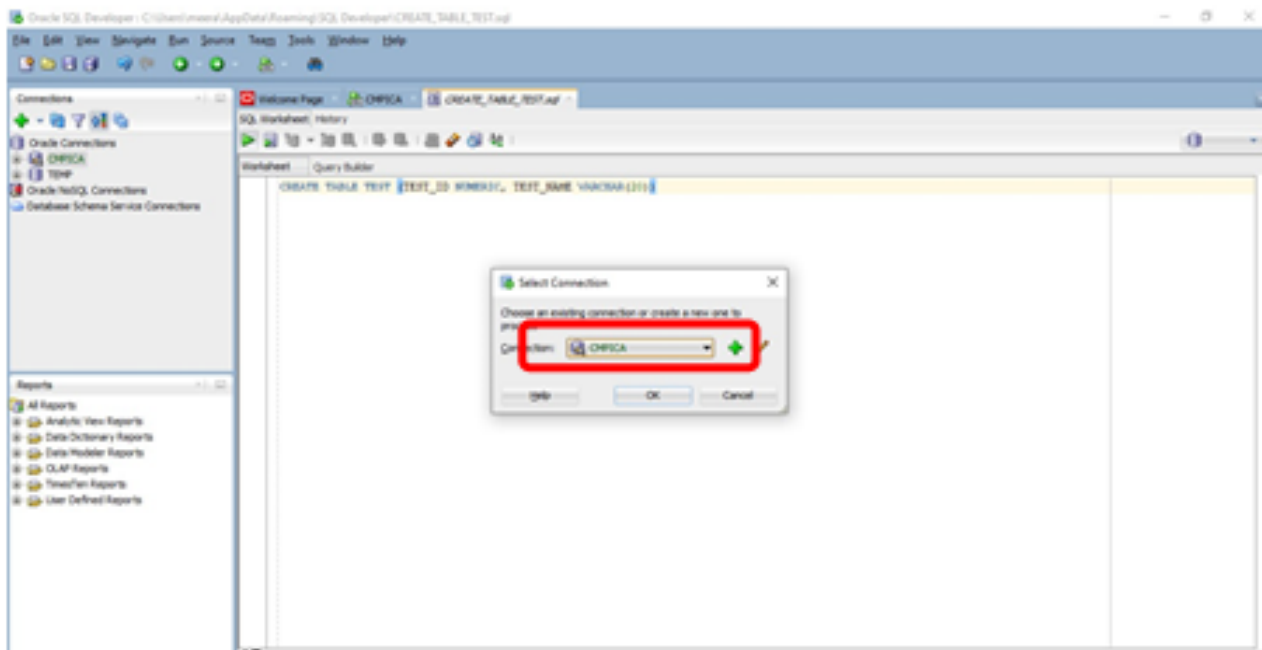
Click OK.

Step 7 Write your query and Run your query (execute) using  icon **OR** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

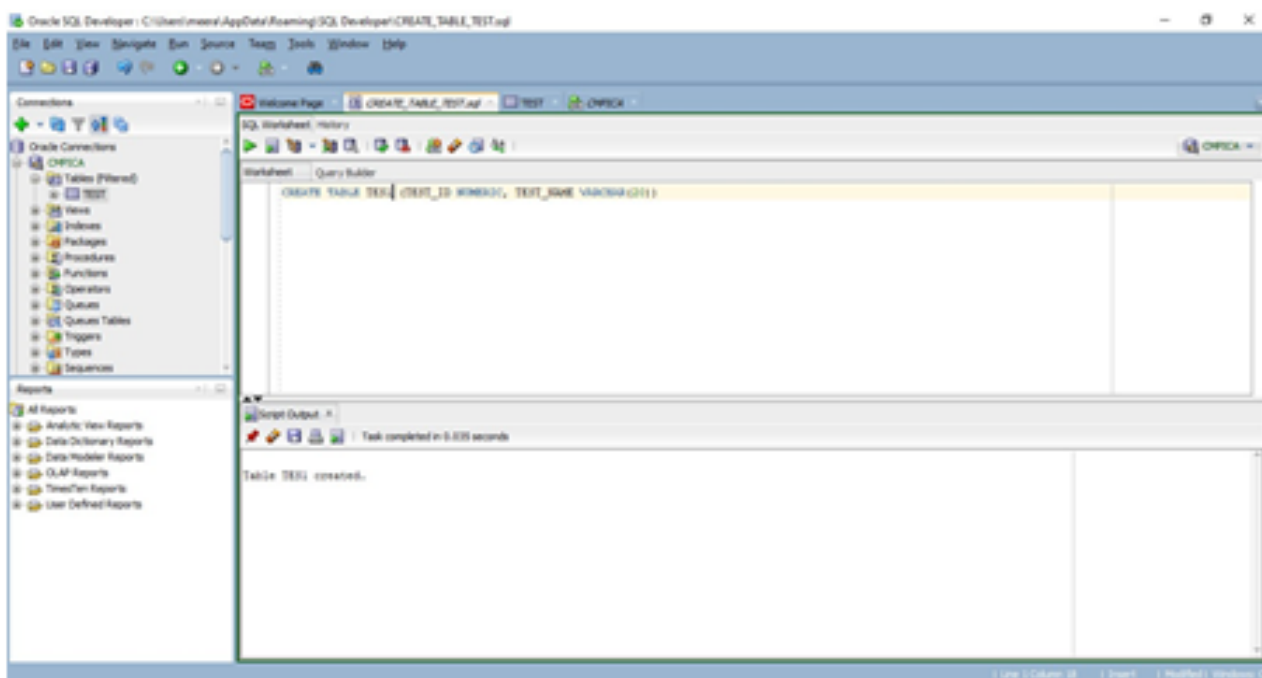


Select the appropriate connection and press OK.

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Observe execution result in bottom panel.

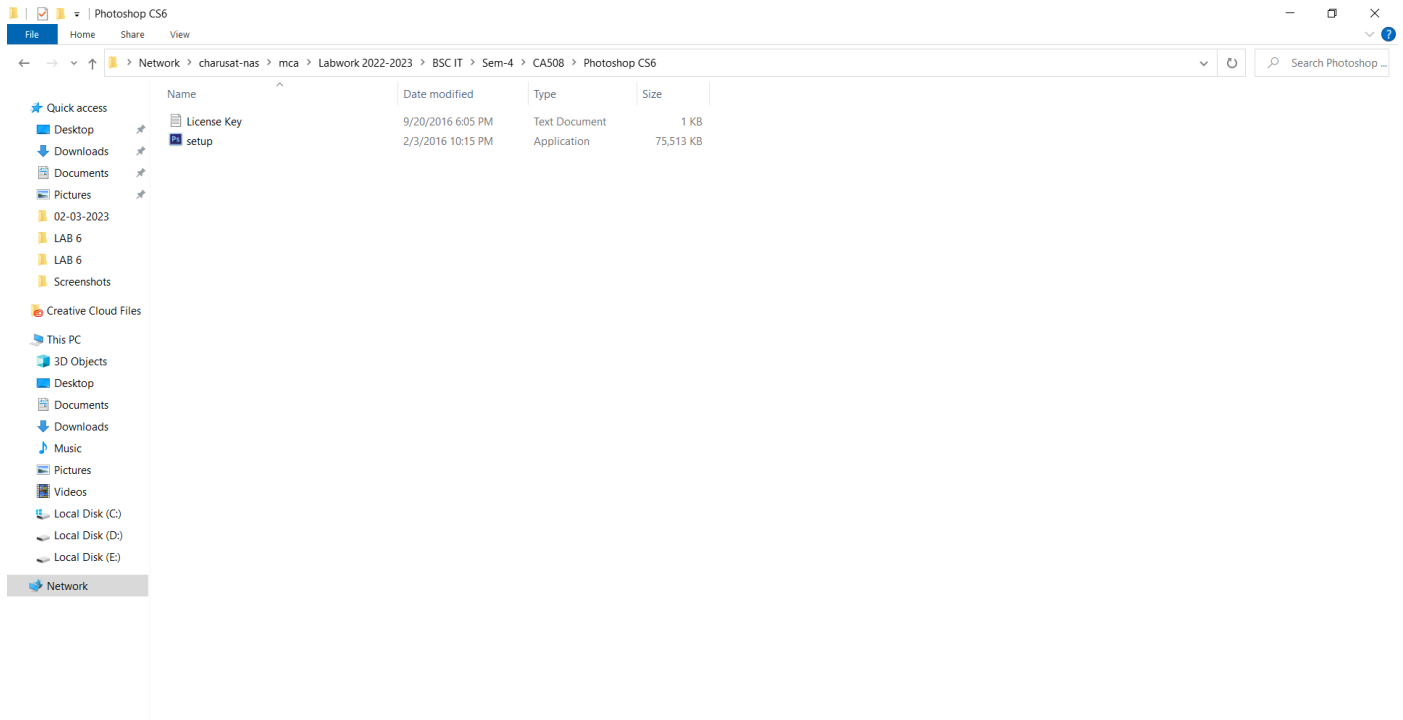


Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.



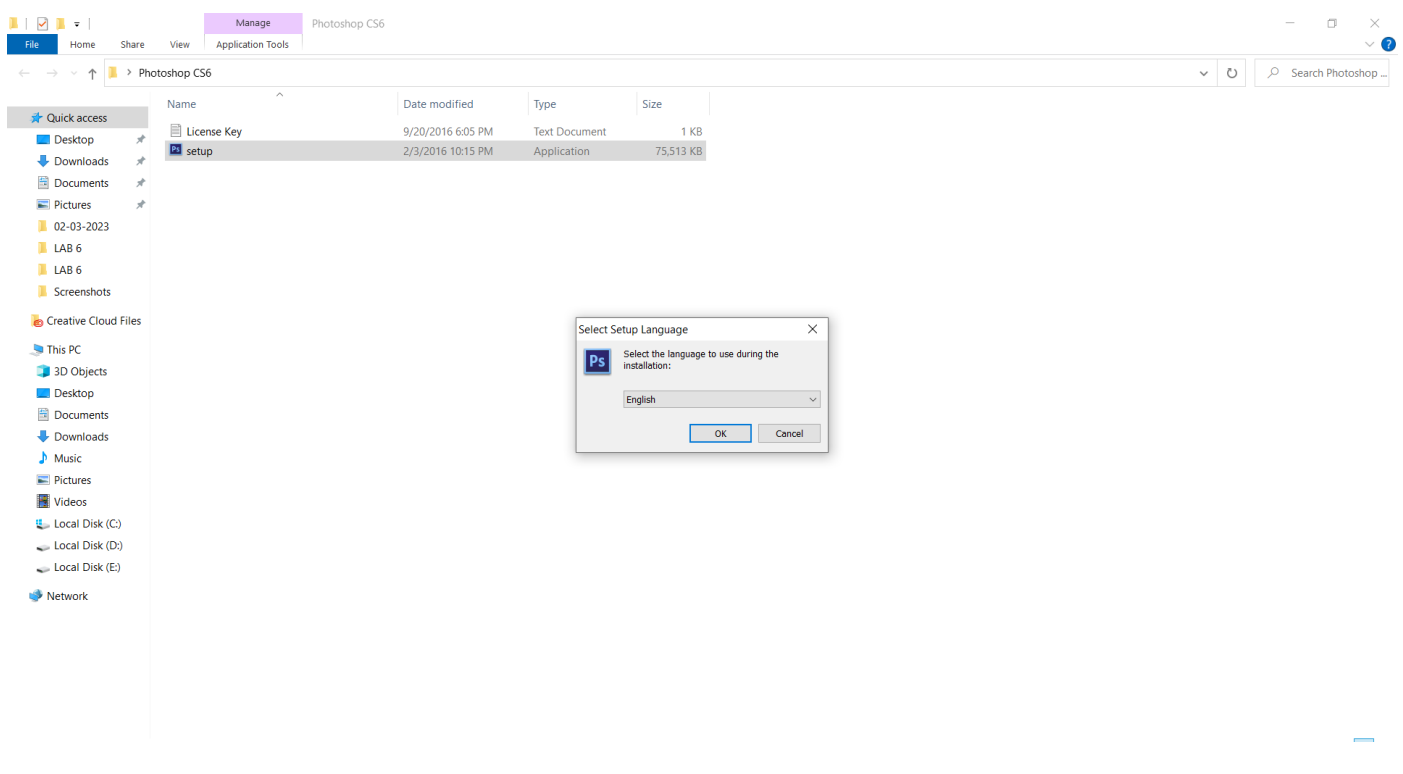
PHOTOSHOP

Download the setup file from <https://www.adobe.com> website.



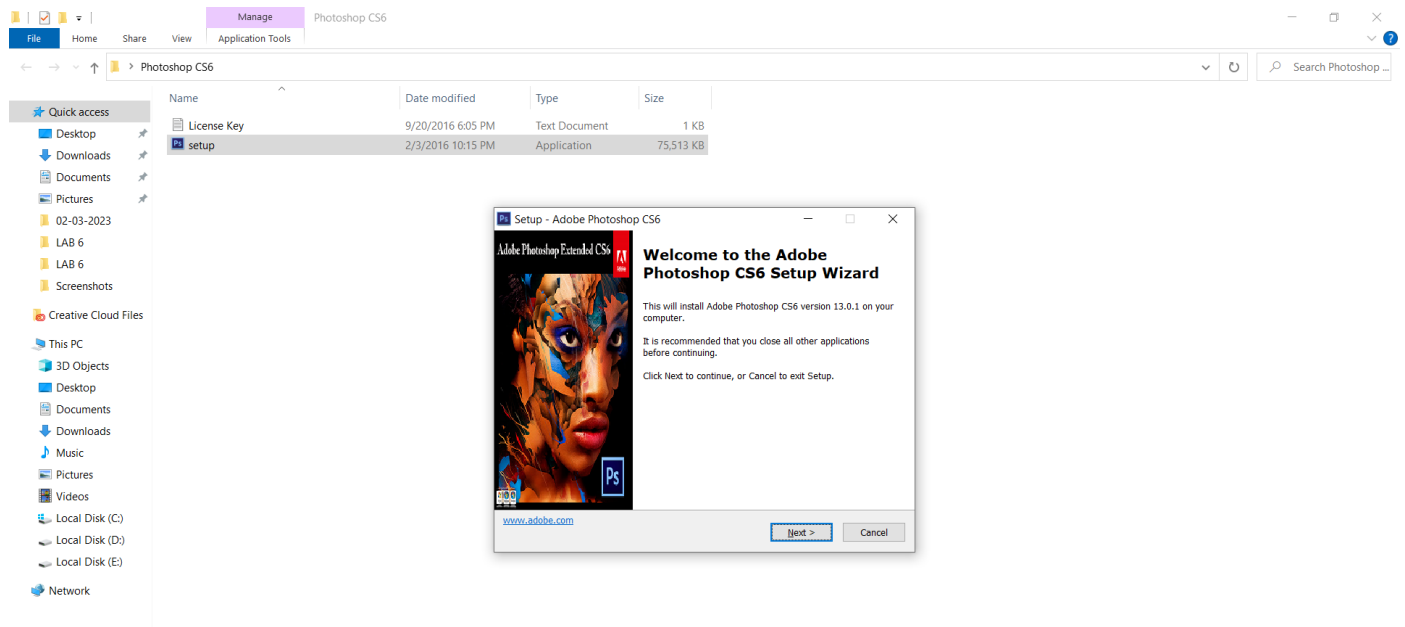
Double click on setup file

S

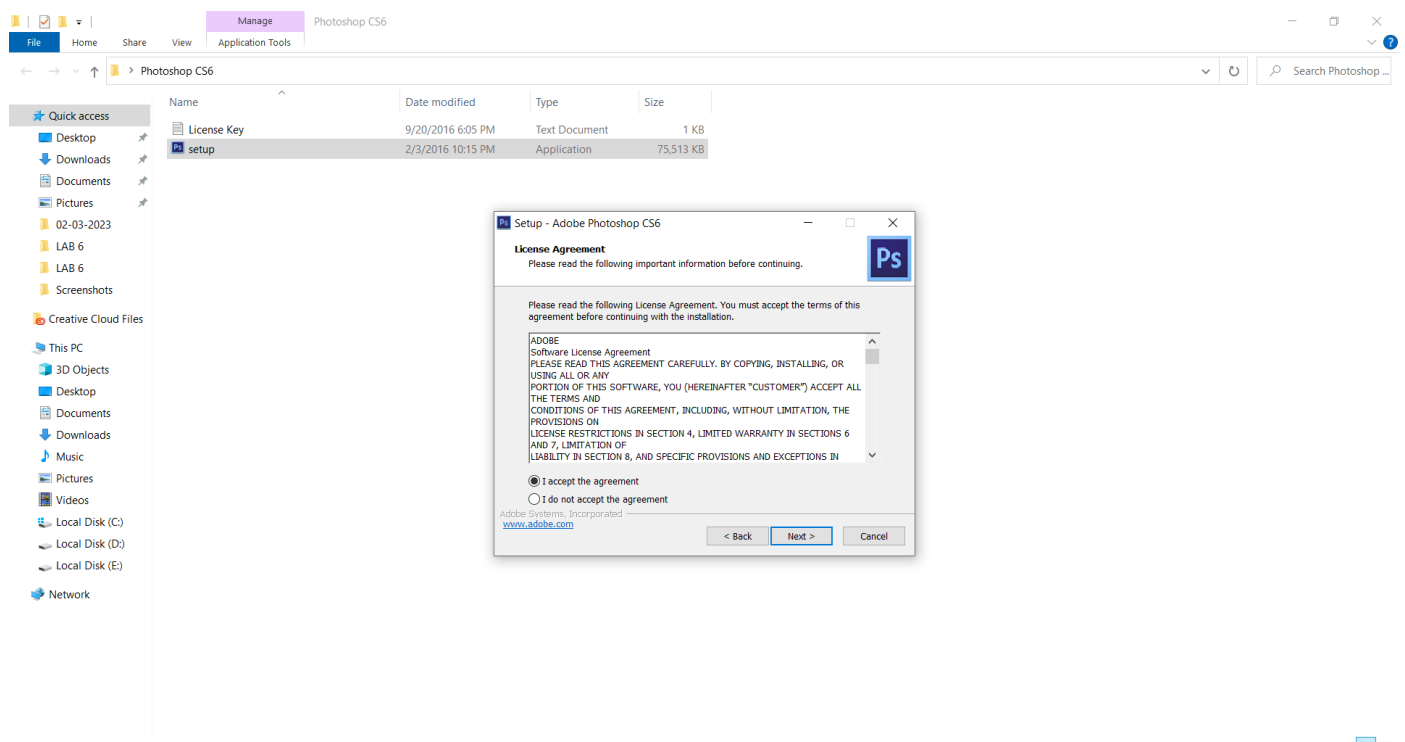


Tool Based Manual || B.SC (IT) || SEM IV || CA508-Graphics Designing

It will show this type of window to select language so select language and click on OK button



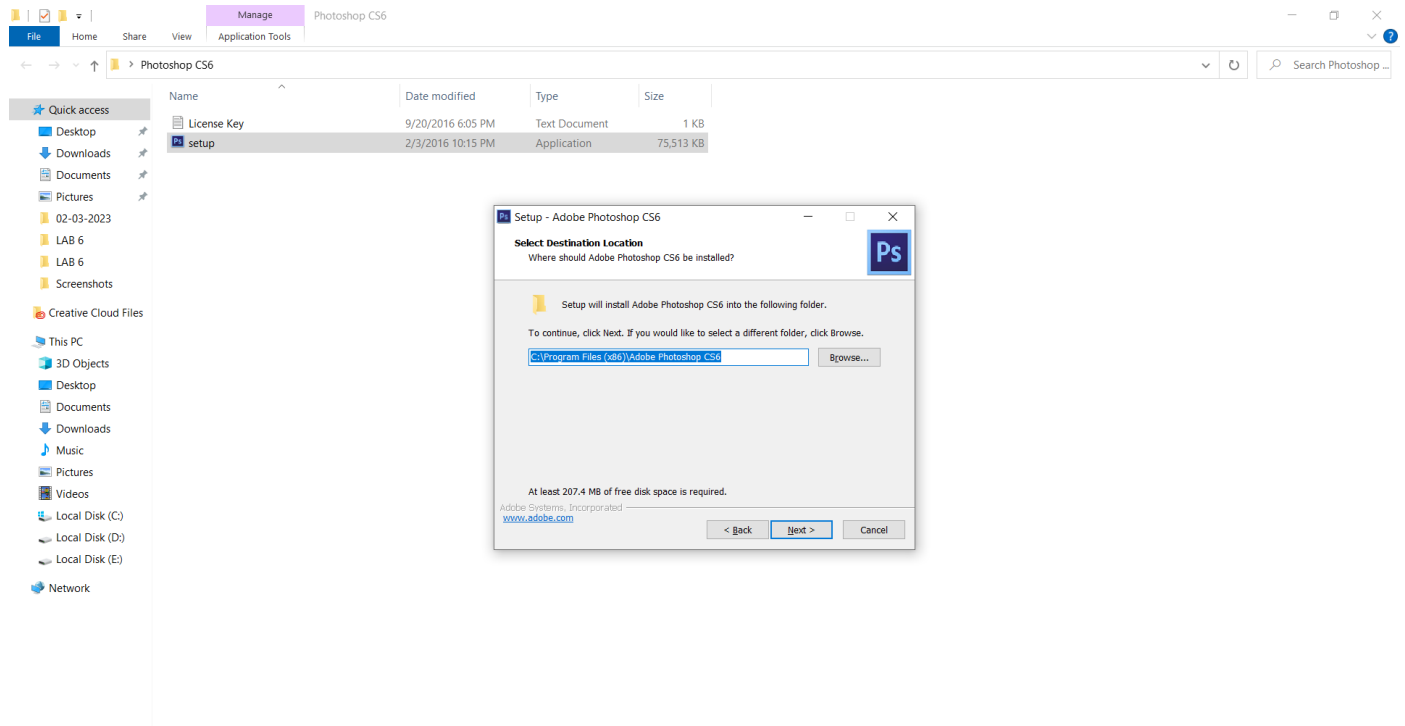
Click on the Next Button.



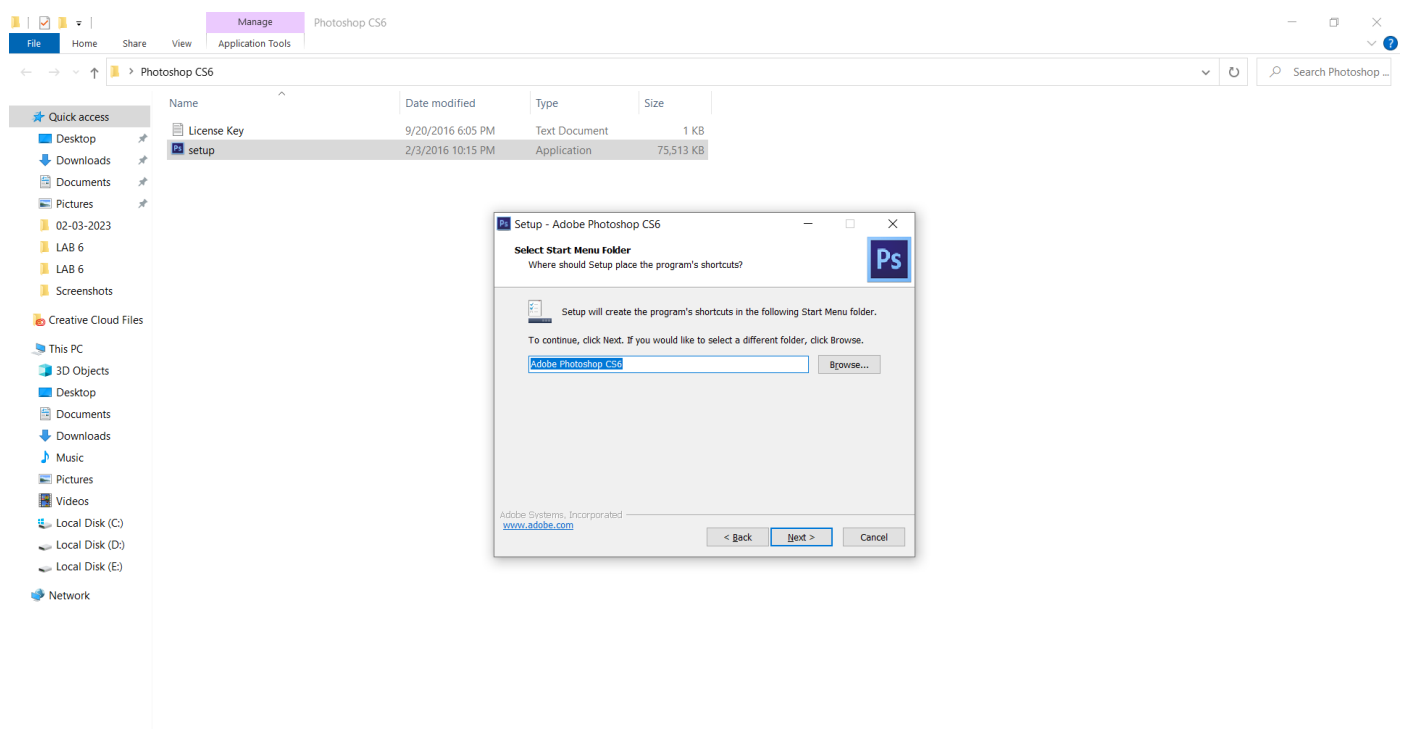


Tool Based Manual || B.SC (IT) || SEM IV || CA508-Graphics Designing

Select the radio button I accept the agreement and then click on next button.

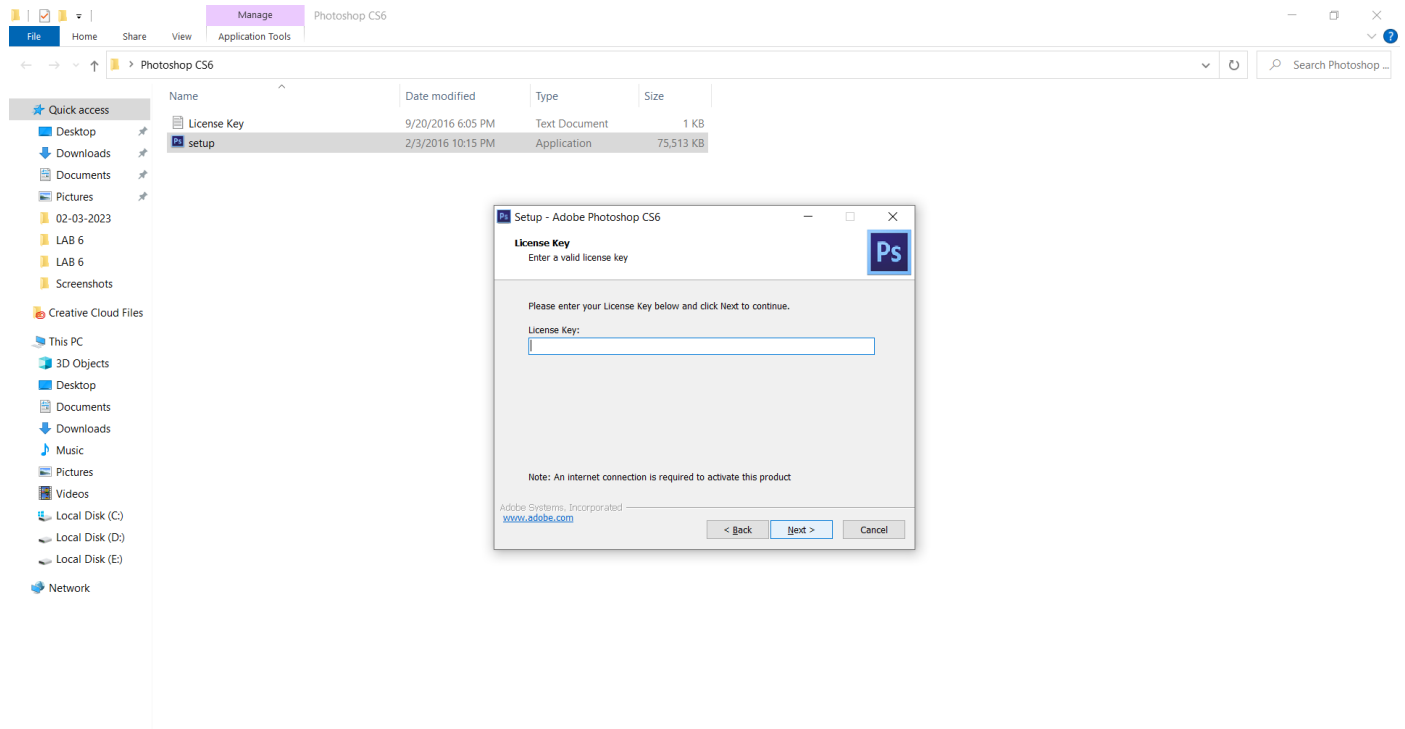


Select the path where you want to install it and click on the next button.



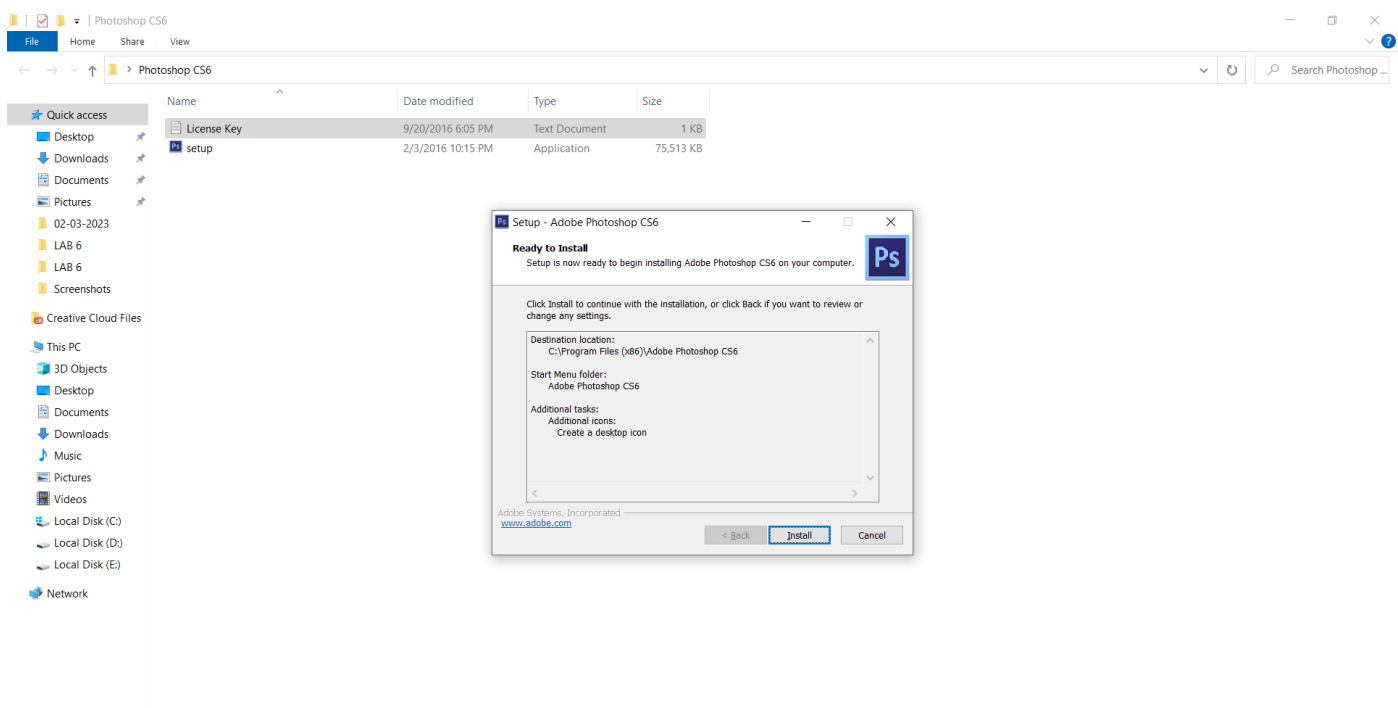
Tool Based Manual || B.SC (IT) || SEM IV || CA508-Graphics Designing

Click on next button after giving any alias name to the software.



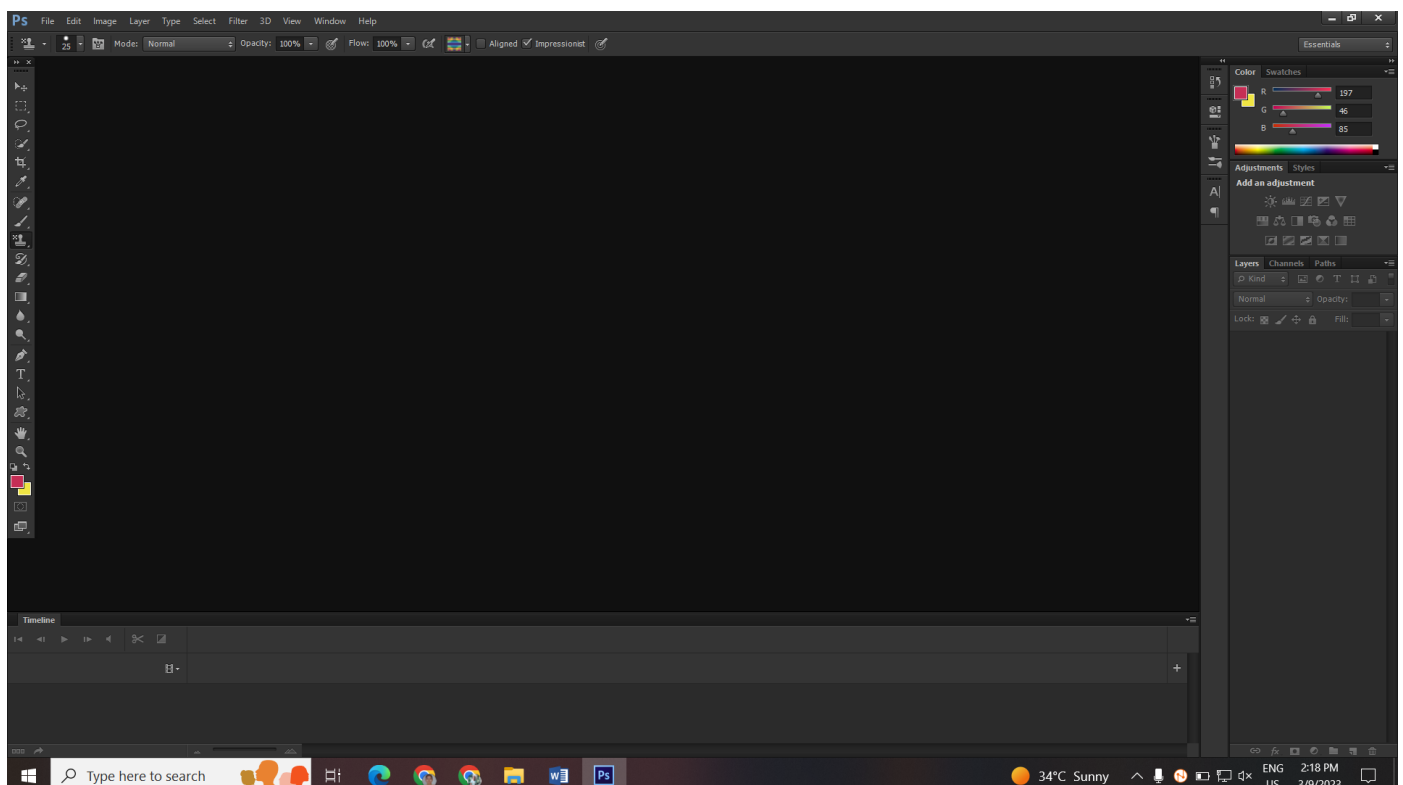
It will ask you to enter license key. So in the setup file you will find one text file which in which there is one code available. Copy that code and paste here and click on the next button.

Tool Based Manual || B.SC (IT) || SEM IV || CA508-Graphics Designing



After entering license key successfully you will find this window and click on install button and it will install the software successfully.

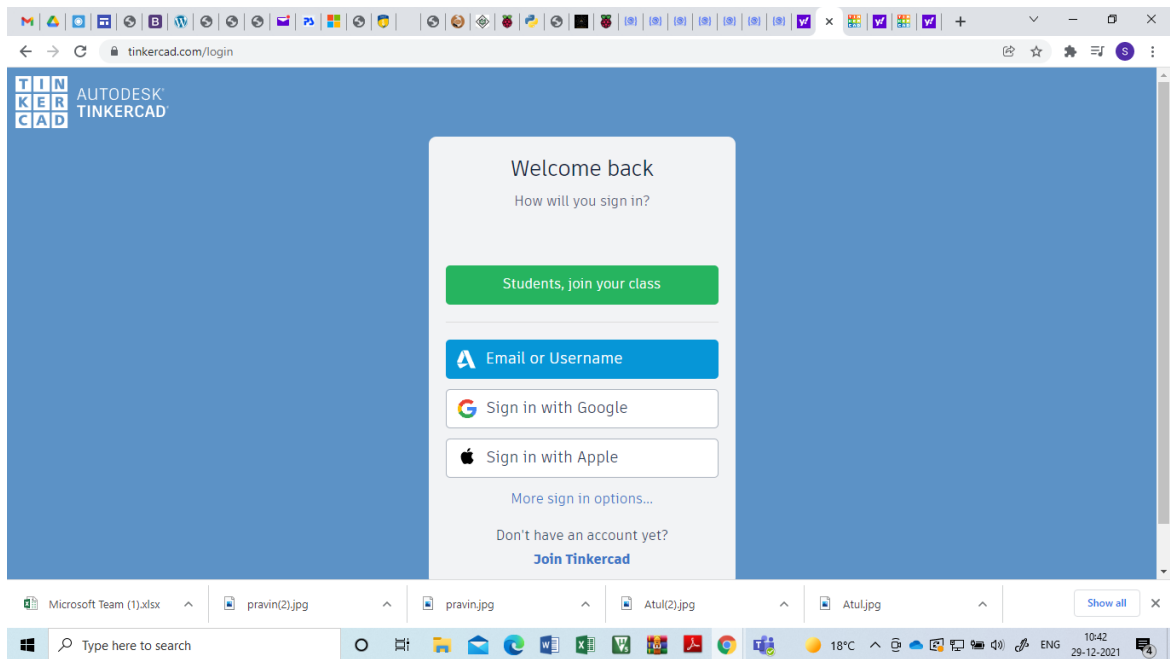
On the desktop you will find the icon of adobe Photoshop just double click on that and you will be able to open the workspace of the Photoshop.



Simulation Using Tinker cad Tool.

Getting Started with Tinkercad

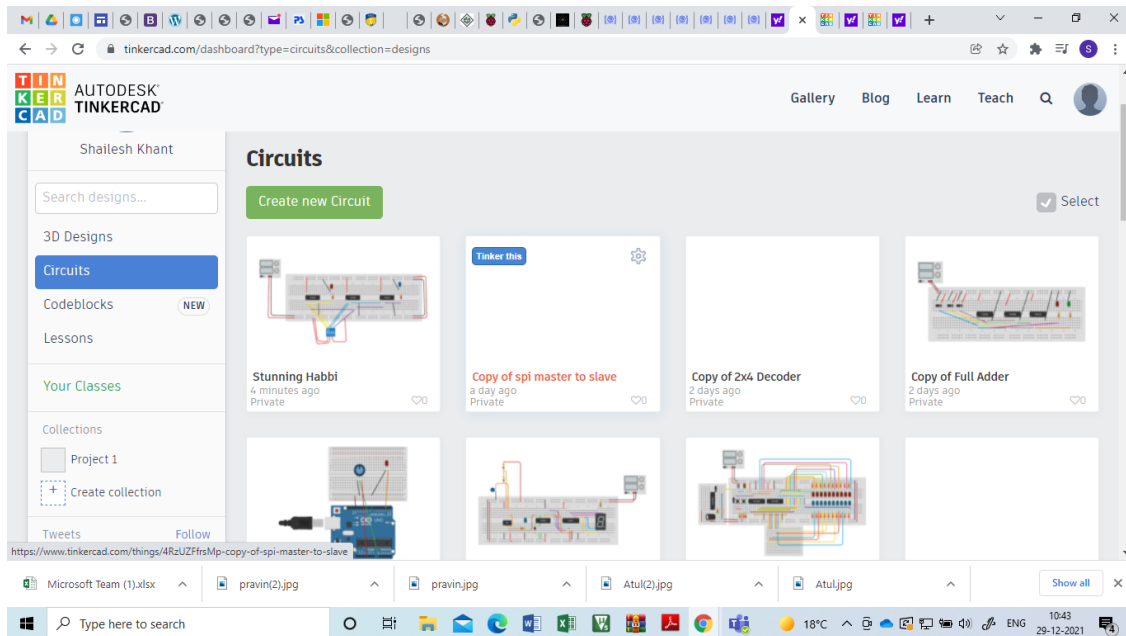
1. Open www.tinkercad.com
2. Login using Google or Facebook account.



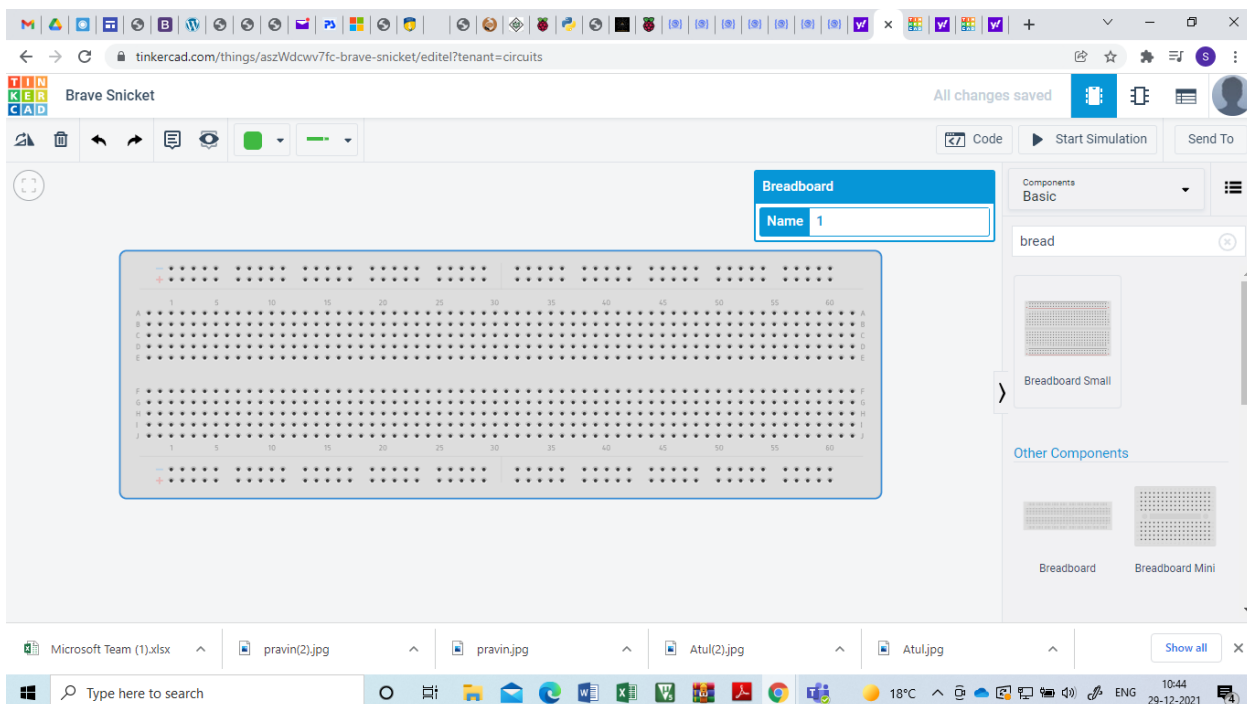
3. Click on new circuit to design new circuit based on electronics hardware.

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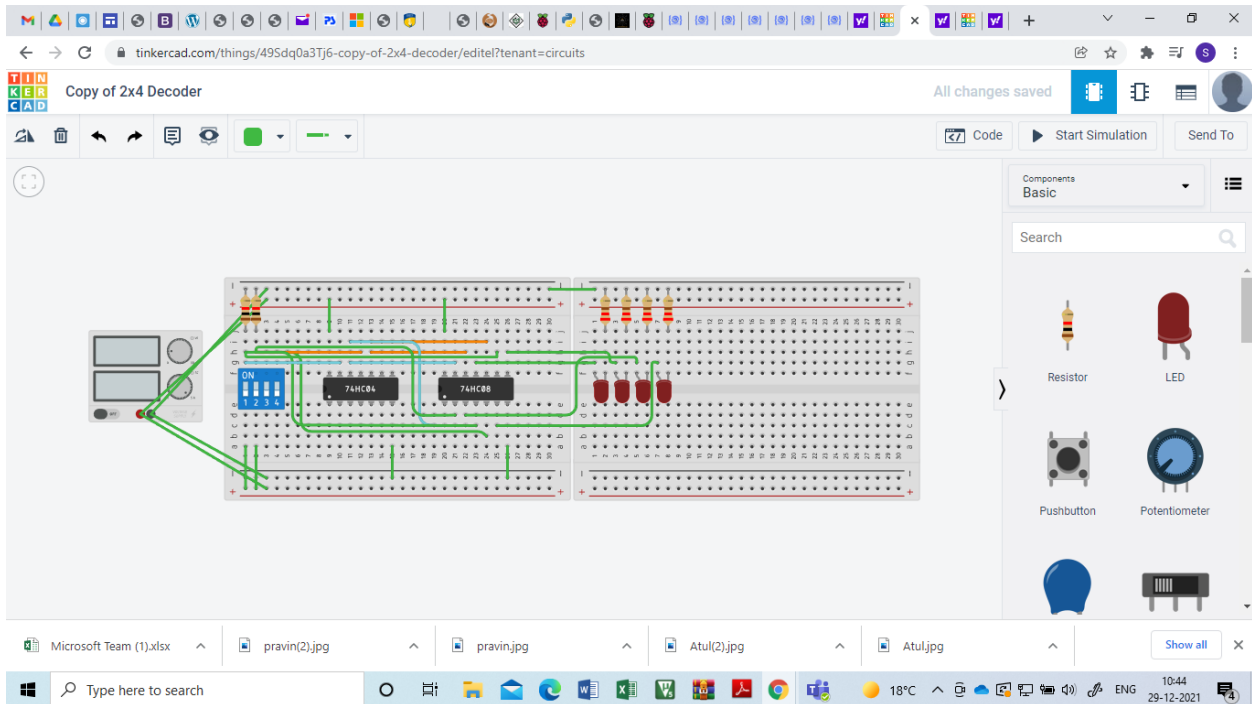
CA615: Embedded Programming using Raspberry PI



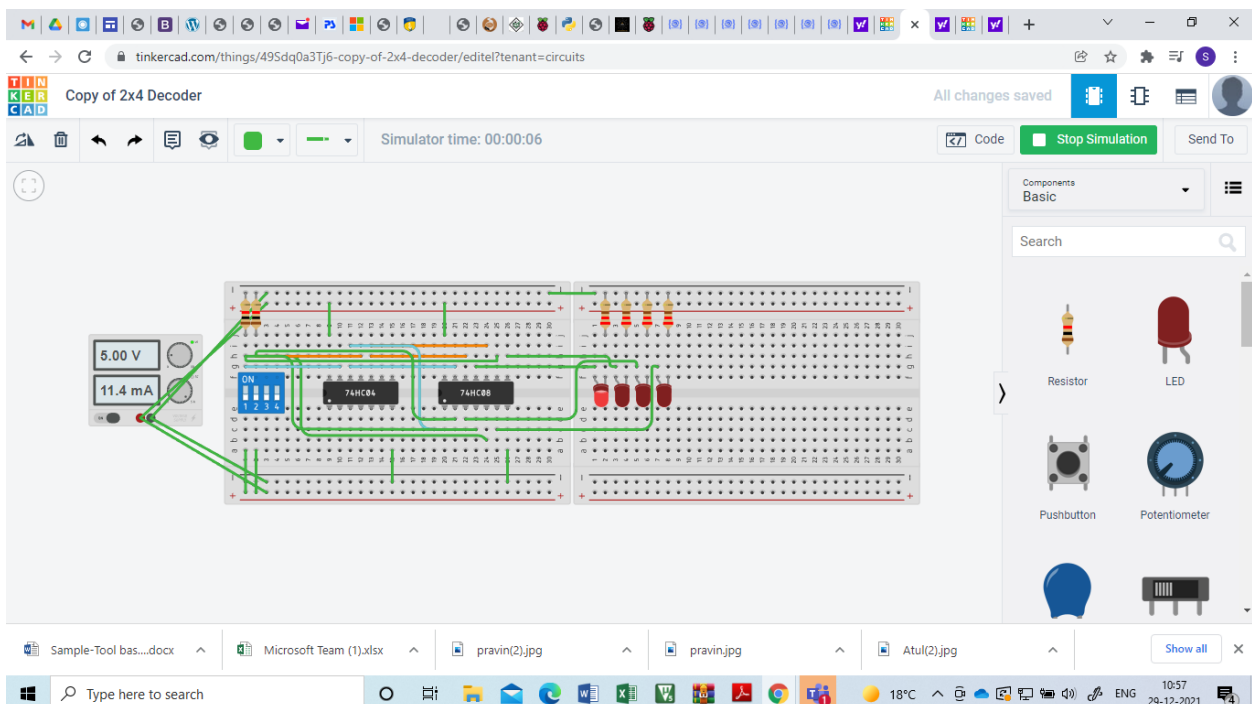
4. Once new circuit is opened then find out the desired components and placed on board.



5. Design whole circuit and check out proper connections.

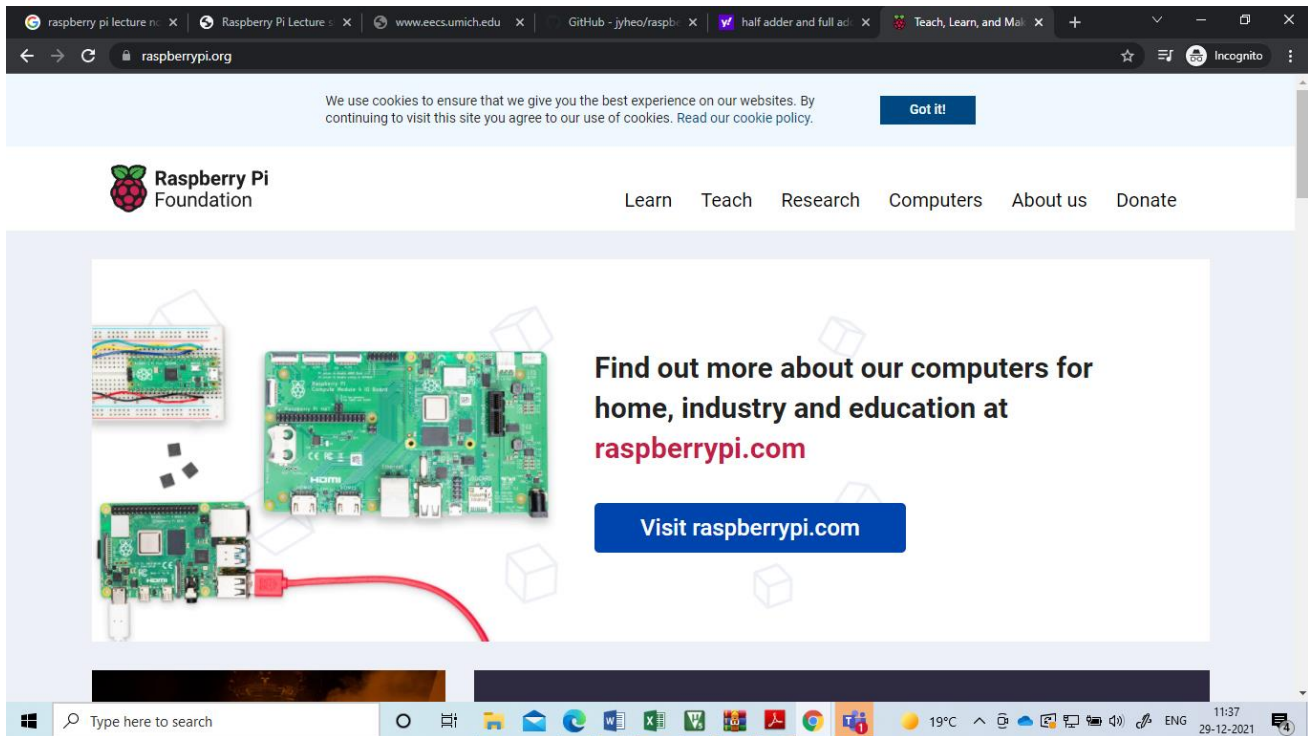


6. Click on start simulation to check the output.



Working with Raspberry PI

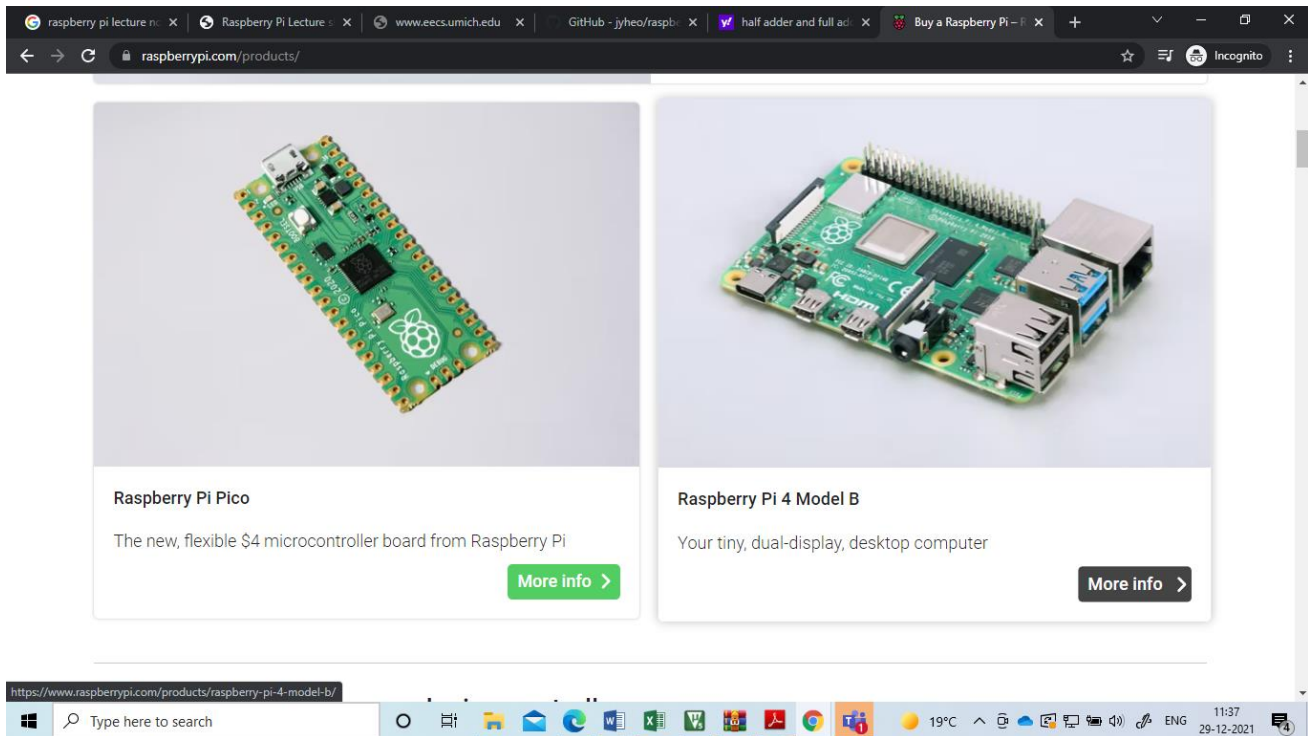
1. Open www.raspberrypi.org



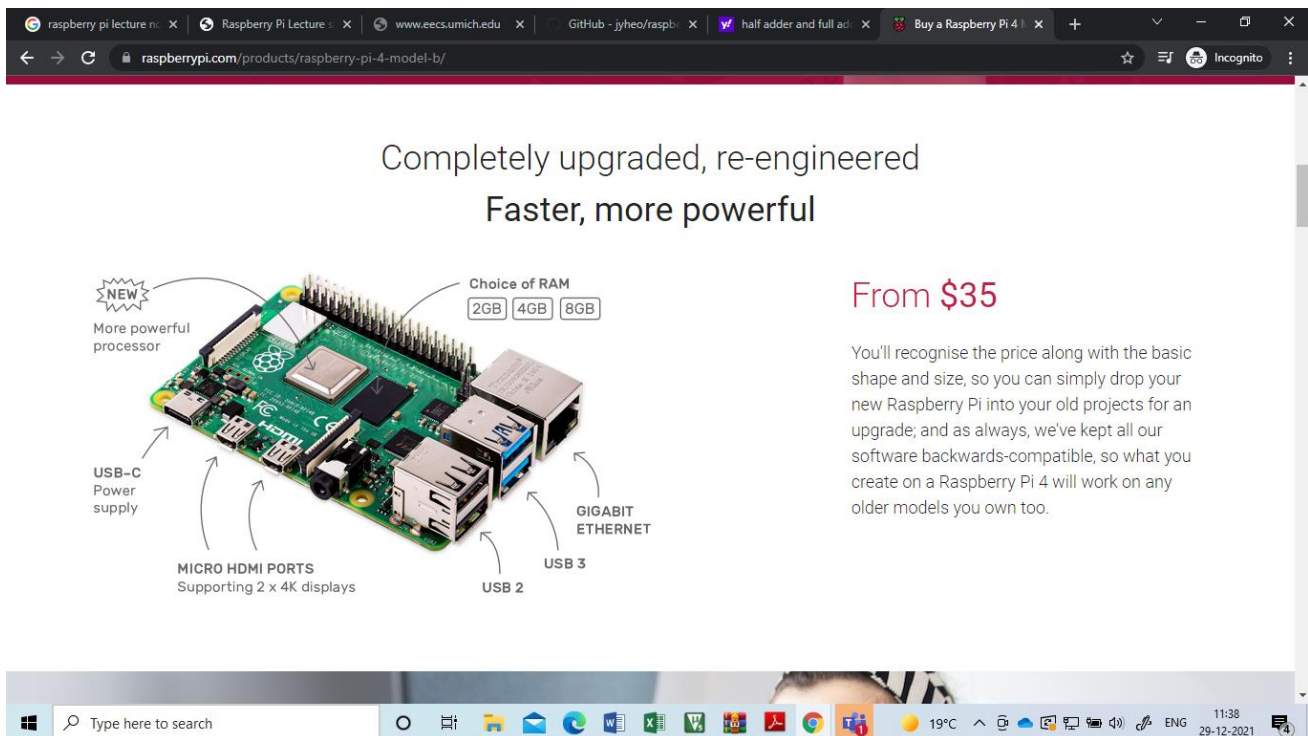
2. Go to product downloads and select hardware.

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CA615: Embedded Programming using Raspberry PI



3. Select Raspberry Pi 4



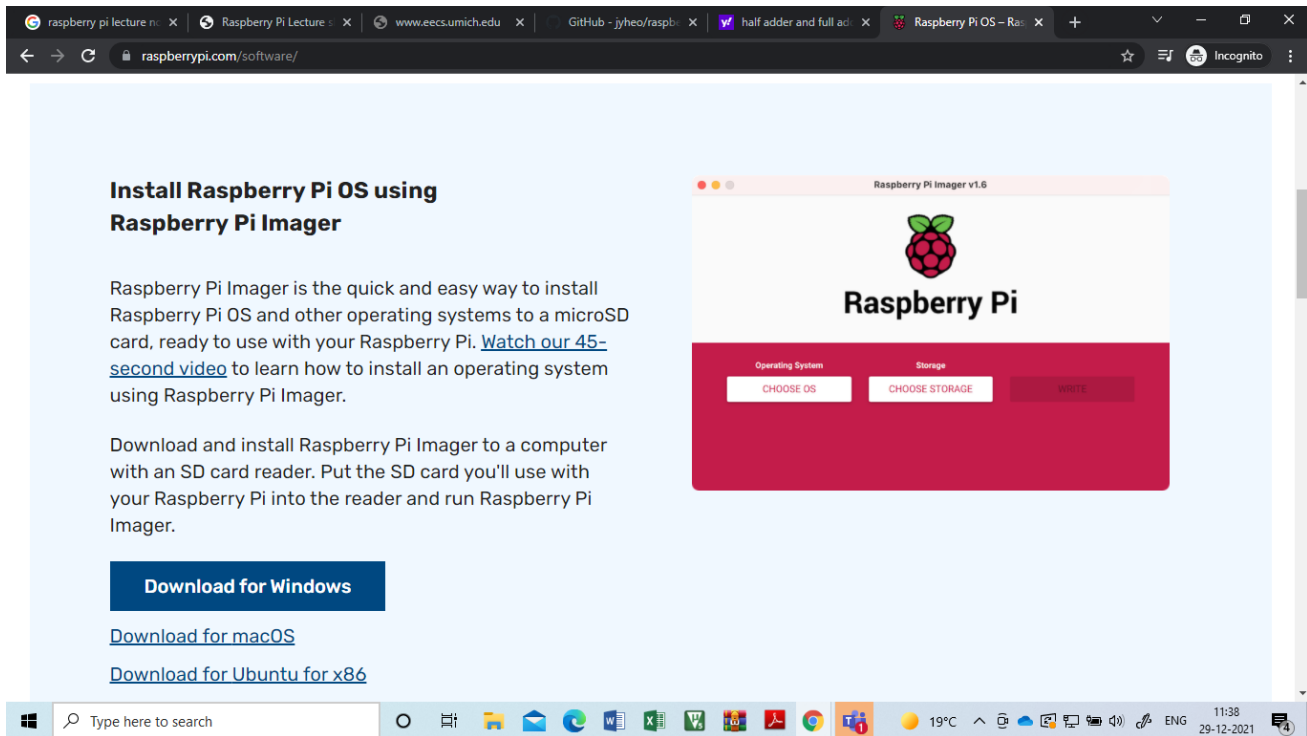
4. Once it is with you then install its OS.



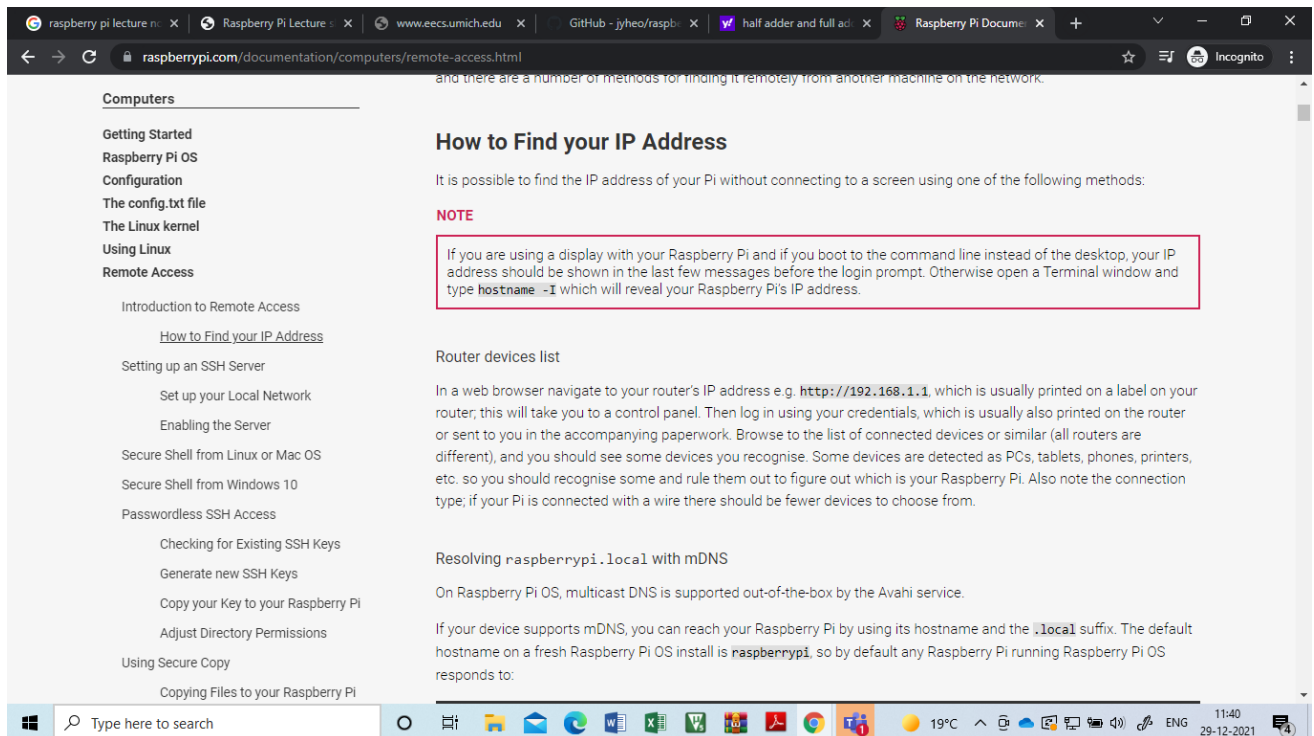
5. Download Raspberry PI OS for windows or linux for operating system in which you are working with.

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CA615: Embedded Programming using Raspberry Pi



6. Install OS on memory chip and connect keyboard , mouse and monitor..
7. Find out IP address of PI.



8. Access your PI remotely using SSH.

The screenshot shows the Raspberry Pi documentation website in a web browser. The left sidebar lists various topics under 'Computers', with 'Remote Access' selected. The main content area is titled 'Set up your Local Network' and includes a 'NOTE' box stating: 'You will need to note down the IP address of your Raspberry Pi in order to connect to it later. Using the `ifconfig` command will display information about the current network status, including the IP address, or you can use `hostname -I` to display the IP addresses associated with the device.' Below this, the 'Enabling the Server' section explains that the SSH server is disabled by default and provides a 4-step guide: 1. Launch Raspberry Pi Configuration from the Preferences menu; 2. Navigate to the Interfaces tab; 3. Select Enabled next to SSH; 4. Click OK. An alternative method using the `raspi-config` application is also shown with two steps: 1. Enter `sudo raspi-config` in a terminal window; 2. Select Interfacing Options.

9. Access your PI remotely using VNC Viewer.

The screenshot shows a Windows desktop environment. In the background, a Microsoft Word document titled 'CA615_EPR_manual.docx' is open. Overlaid on the Word document is the VNC Viewer application window. The 'Properties' dialog box is open, showing the 'General' tab. The 'VNC Server' field is empty, with a placeholder text 'IP address or hostname'. The 'Name' field is empty, with a placeholder text 'Friendly identifier'. The 'Labels' section has a text box for entering labels. The 'Security' section has a dropdown menu set to 'Let VNC Server choose', and two checkboxes are checked: 'Authenticate using single sign-on (SSO) if possible' and 'Authenticate using a smartcard or certificate store if possible'. The 'Privacy' section has a checkbox for 'Update desktop preview automatically' which is also checked. The 'OK' and 'Cancel' buttons are at the bottom of the dialog box.

10. Start working with raspberry PI projects.
11. Use python for programming hardware using raspberry PI.
12. Install the packages as needed by the projects.

Welcome to Spyder (Python Interpreter)

The screenshot displays the Spyder Python IDE interface. The left sidebar shows a file explorer with a project structure including 'plugin.py', 'Plots', 'chart_plot_example.py', and 'plugin.py - ipythonconsole'. The central pane shows the code editor with the 'plugin.py' file open, displaying Python code for a plugin. The right pane shows the 'Variable Explorer' with a table of variables and their values. Below the variable explorer, there is a 3D plot of a terrain model and a polar plot.

Name	Type	Size	Value
bool	bool	1	True
data	Array of str128	(3, 3)	ndarray object of numpy module
datetime_object	datetime	1	2021-04-14 17:35:14.687085
df	DataFrame	(2, 2)	Column names: Col1, Col2
filename	str	53	/Users/Documents/spyder/spyder/tests/test_dont_use.py
li	list	5	['abcd', 745, 2.23, 'efgh', 70.2]
myset	set	3	{'2', '1', '3'}
r	float	1	6.46567886443
t	tuple	5	('abcd', 745, 2.23, 'efgh', 70.2)
tinylst	list	2	[123, 'efgh']
x	float64	1	1.1235123899439

Spyder is a powerful scientific environment written in Python, for Python, and designed by and for scientists, engineers and data analysts. It features a unique combination of the advanced editing, analysis, debugging, and profiling functionality of a comprehensive development tool with the data exploration, interactive execution, deep inspection, and beautiful visualization capabilities of a scientific package.

CA615: Embedded Programming using Raspberry PI

List of Practical

Sr No	Title of the Experiment
1.	Implementing digital components and circuits using Tinker cad.
2.	Connecting and designing various circuits using Tinker cad.
3.	Introduction and installation of Raspberry PI Board and its OS.
4.	Connecting Raspberry Pi virtually using VNC Viewer, SSH and Remote desktop application.
5.	Introduction to Python for Raspberry PI.
6.	Python Programming for Raspberry PI: Basics.
7.	Python Programming for Raspberry PI: Operators, Decision making and loops.
8.	Python Programming for Raspberry PI: Numbers and strings.
9.	Python Programming for Raspberry PI: List, Tuple and dictionary.
10.	Python Programming for Raspberry PI: Functions.
11.	Interfacing Raspberry PI with sensors.
12.	Interfacing Raspberry PI with camera modules.